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1

Matrix Algebra

Learning Objectives

After working through this chapter you will be able to:

- read and write matrices with index notation, and say what the *size* of a matrix means;
- add, scale, transpose and multiply matrices, and check sizes *before* computing;
- explain, with examples, why $AB \neq BA$ in general and why cancellation can fail;
- recognize diagonal, triangular, symmetric and skew-symmetric matrices;
- interpret a matrix as a *transformation of the plane*, and as a model of trusses, circuits, spring systems and robot arms.

1.1 Introduction to Matrices

Imagine a small electronics company that sells three products: **Product A (laptops), Product B (tablets), and Product C (smartphones)**. The marketing team wants to analyze sales over the last four months to decide next quarter's inventory. How can they organize this data so that it is easy to analyze? They can arrange it in a rectangular grid, which we call a **matrix** (Figure 1.1). Each row represents a month and each column represents a product:

$$S = \begin{array}{c} \\ \text{Month 1} \\ \text{Month 2} \\ \text{Month 3} \\ \text{Month 4} \end{array} \begin{array}{c} \begin{array}{ccc} \text{A} & \text{B} & \text{C} \end{array} \\ \begin{bmatrix} 100 & 150 & 200 \\ 120 & 140 & 180 \\ 110 & 160 & 190 \\ 130 & 170 & 210 \end{bmatrix} \end{array}$$

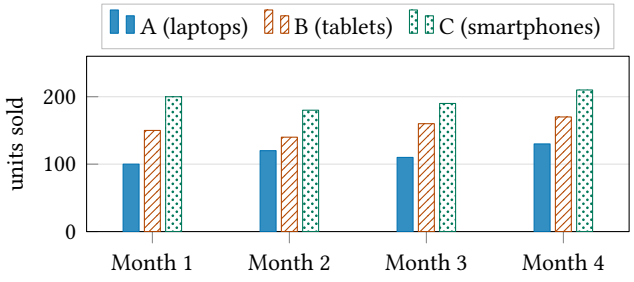
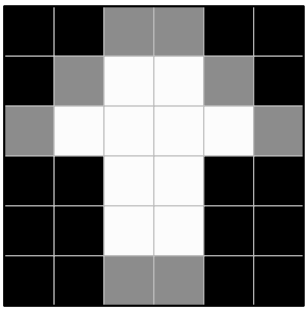


Figure 1.1: The sales matrix S as a bar chart. Each *group* of bars is one row of S (a month); each *bar style* is one column (a product). Product C (dotted) is the highest in every month.

Using this matrix, we can answer business questions with simple operations:

- **Total sales in Month 3:** add the third row: $110 + 160 + 190 = 460$ units.
- **Top-selling product overall:** add each column. The column sums are 460 (A), 620 (B) and 780 (C), so Product C (smartphones) sells the most: $200 + 180 + 190 + 210 = 780$ units.

Organizing data in matrix form lets us use **linear algebra** to run complex operations, expose structure, and compute solutions efficiently. Matrices are not only tables of business data. A digital image is also a matrix: one brightness value per pixel (Figure 1.2). Brightening the image multiplies the matrix by a number, and blending two photographs adds two matrices; we define both operations in Section 1.2.



(a) Visual image: a 6×6 grid of square pixels.

$$\begin{bmatrix} 0 & 0 & 5 & 5 & 0 & 0 \\ 0 & 5 & 9 & 9 & 5 & 0 \\ 5 & 9 & 9 & 9 & 9 & 5 \\ 0 & 0 & 9 & 9 & 0 & 0 \\ 0 & 0 & 9 & 9 & 0 & 0 \\ 0 & 0 & 5 & 5 & 0 & 0 \end{bmatrix}$$

(b) Corresponding matrix stored in memory.

Figure 1.2: A digital image is a grid of pixels, where each pixel square holds a single brightness value (0 = black, 5 = mid-gray edge, 9 = white). The computer stores this grid directly as a matrix.

1.1.1 Definition and Notation

Definition

A **matrix** is a rectangular array of numbers (or functions) arranged in rows and columns. A matrix with m rows and n columns has **size** (also called *dimension* or *order*) $m \times n$, read "m by n". **Rows are always counted first.**
Capital letters (A, B, C) denote matrices, and lowercase letters with two subscripts (a_{ij}) denote entries. The entry in row i and column j of A is a_{ij}. In compact form, $A = [a_{ij}]_{m \times n}$.

A matrix with one row is a **row matrix**; a matrix with one column is a **column matrix**. An $m \times n$ matrix A looks like this (Figure 1.3):

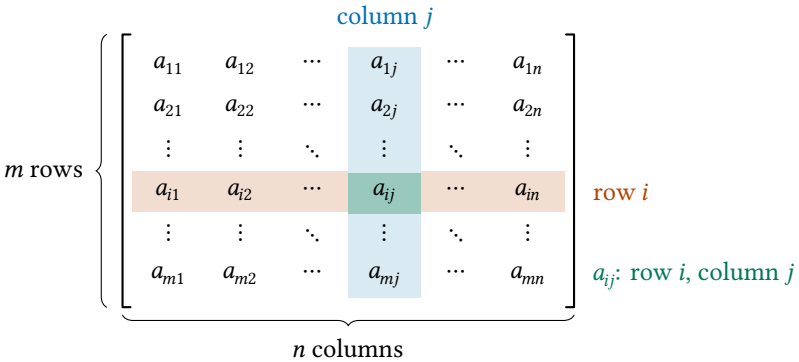


Figure 1.3: Anatomy of an $m \times n$ matrix. Row i (shaded orange) and column j (shaded blue) cross at the single entry a_{ij} (shaded green).

- The entry a_{ij} lies at the intersection of the i-th row and the j-th column.
- The ***i*-th row** is the row matrix $\begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix}$.
- The ***j*-th column** is the column matrix $\begin{bmatrix} a_{1j} & a_{2j} & \cdots & a_{mj} \end{bmatrix}^T$ (the symbol ^{T} is explained in Section 1.2.3).

Example: Matrix Notation

Consider the 2×3 matrix A:

$$A = \begin{bmatrix} 1 & -3 & 5 \\ 0 & 4 & 2 \end{bmatrix}$$

Here $a_{12} = -3$ and $a_{23} = 2$. There is no entry a_{32}, because A has only two rows.

Common mistake. Two matrices that contain the same numbers are *not* equal if their sizes differ. The row $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ is 1×3, the column $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$ is 3×1, and they are different objects.

Quick Check 1

1. A matrix has 12 entries. List every possible size it could have.
2. In the sales matrix S, what does the entry s_{23} mean, and what is its value?
3. Is $\begin{bmatrix} 1 & 2 \end{bmatrix}$ equal to $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$? Explain.

1.2 Matrix Algebra: Basic Operations

Unlike real numbers, matrices can be added, subtracted and multiplied only when their *sizes fit together*. Division is not defined directly; the matrix inverse plays its role (Chapter 2).

1.2.1 Equality, Addition, and Subtraction

Definition

Two matrices $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{p \times q}$ are **equal** ($A = B$) if and only if they have the same size ($m = p$ and $n = q$) and $a_{ij} = b_{ij}$ for all i, j.
The **sum** $A + B$ and **difference** $A - B$ are defined entry by entry, and *only* for matrices of the same size:

$$A \pm B = [a_{ij} \pm b_{ij}]_{m \times n}.$$

Example: Addition and Subtraction

Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$:

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \qquad A - B = \begin{bmatrix} -4 & -4 \\ -4 & -4 \end{bmatrix}.$$

If A and B were the sales matrices of two shops, $A + B$ would be their combined sales.

Common mistake. A 2×3 matrix and a 3×2 matrix *cannot* be added, even though they contain the same number of entries. Always compare sizes first.

1.2.2 Scalar Multiplication

Definition

Let $A = [a_{ij}]_{m\times n}$ be a matrix and $k \in \mathbb{R}$ a scalar. The **scalar product** kA multiplies every entry of A by k:

$$kA = [k \cdot a_{ij}]_{m\times n}.$$

An expression such as $kA + lB$ is called a **linear combination** of A and B.

Example: A Linear Combination

Let A and B be as in the addition and subtraction example,

$$3A - 2B = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} - \begin{bmatrix} 10 & 12 \\ 14 & 16 \end{bmatrix} = \begin{bmatrix} -7 & -6 \\ -5 & -4 \end{bmatrix}.$$

Intuition. Let A and B be the matrices of two images like the one in Figure 1.2. Then $2A$ is the same picture with doubled brightness values (anything brighter than white is clipped to white), and $\frac{1}{2}A + \frac{1}{2}B$ is a 50 % blend of the two pictures.

1.2.3 The Transpose

Definition

The **transpose** of an $m\times n$ matrix $A = [a_{ij}]$ is the $n\times m$ matrix A^T whose (i, j) entry is a_{ji}. In words: **rows become columns**.

For matrices of compatible sizes and a scalar k:

$$(A^T)^T = A, \qquad (A + B)^T = A^T + B^T, \qquad (kA)^T = kA^T.$$

The rule for products, $(AB)^T = B^T A^T$, is proved in Section 1.3.1.

Example: Transpose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad\Longrightarrow\quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

Row 1 of A (orange) is column 1 of A^T, and row 2 (blue) is column 2. A 2×3 matrix becomes a 3×2 matrix.

1.2.4 Matrix Multiplication

Matrix multiplication is a tool for combining two connected processes into a single direct operation, where the output categories of the first stage match the input categories of the second. We do not define matrix multiplication entry by entry (grid cell by grid cell). That operation exists, but it multiplies numbers in matching positions and leaves them completely isolated: it does not mix information across stages or combine separate contributions into a single total. Instead, matrix multiplication uses

a "row-by-column" rule—multiplying matching pairs across a row and a column and adding them together—to collapse the shared middle link and calculate how initial inputs directly produce final outputs.

Definition

Let A be an $m \times n$ matrix and B an $n \times p$ matrix. The product $C = AB$ is the $m \times p$ matrix whose entry c_{ik} is the dot product of the i-th row of A with the k-th column of B:

$$c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk} = a_{i1} b_{1k} + a_{i2} b_{2k} + \cdots + a_{in} b_{nk}.$$

Compatibility rule: AB is defined **if and only if** the number of columns of A equals the number of rows of B.

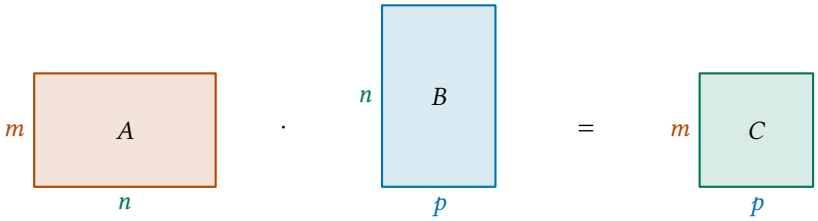


Figure 1.4: Size bookkeeping for $C = AB$. The two *inner* sizes (n, in green) must agree; the *outer* sizes m and p give the size of the product. Note that the width of A equals the height of B.

Intuition. Why is multiplication defined in such a strange way? Suppose the shop from Section 1.1 charges \$900 for a laptop, \$500 for a tablet and \$700 for a smartphone. The revenue in Month 1 is $100 \cdot 900 + 150 \cdot 500 + 200 \cdot 700 = 305{,}000$. This is a "row times column" sum. Doing it for all four months at once gives

$$S\mathbf{p} = \begin{bmatrix} 100 & 150 & 200 \\ 120 & 140 & 180 \\ 110 & 160 & 190 \\ 130 & 170 & 210 \end{bmatrix} \begin{bmatrix} 900 \\ 500 \\ 700 \end{bmatrix} = \begin{bmatrix} 305{,}000 \\ 304{,}000 \\ 312{,}000 \\ 349{,}000 \end{bmatrix}.$$

Matrix multiplication is exactly the bookkeeping for "combine many quantities with many weights".

Visualization: Row-by-Column Multiplication

To compute c_{11} for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, multiply the first row of A by the first column of B:

$$c_{11} = (1 \times 5) + (2 \times 7) = 5 + 14 = 19.$$

Computing all entries yields

$$AB = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}.$$

Another way to read a product

Figure 1.5 shows Falk's scheme for matrix multiplication. It places the matrices A ($m \times n$), B ($n \times p$), and $C = AB$ ($m \times p$) on a grid, highlights the second row of A and the second column of B, and shows how the sum of the products of corresponding elements produces the entry c_{22}. Blue projection lines connect the relevant row and column to C, while red arrows indicate the individual products and their sum.

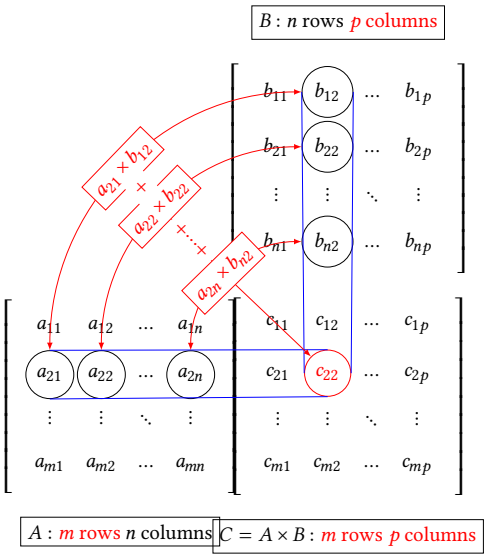


Figure 1.5: Falk's scheme. Put A to the left and B above the product. Every entry of $C = AB$ sits at the crossing of one row of A and one column of B.

1.2.5 Matrices as Transformations of the Plane

Before we look at how matrices transform space, it helps to clarify what a **vector** is.

Algebraically, a vector is simply a specialized, single-column matrix (an $n \times 1$ matrix). While a general matrix is a two-dimensional grid of numbers, a column vector like $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ is a one-dimensional list of numerical components.

Geometrically, a vector is represented by a **directed line segment** (an arrow). Crucially, a vector describes a *displacement*—an instruction to move x units horizontally and y units vertically:

- **Free Vectors:** The initial point (tail) of the arrow can be placed *anywhere* in the plane. As long as two directed segments have the same length and point in the same direction, they represent the exact same vector.
- **Standard Position:** To connect geometry with algebra, we usually place the tail at the **origin** $(0, 0)$. When rooted at the origin, the head of the vector points directly to the point (x, y), making the matrix entries $\begin{bmatrix} x \\ y \end{bmatrix}$ double as coordinates.

Connecting Geometry and Algebra: These perspectives are two sides of the same coin:

- The **directed line segment** gives us spatial intuition (length and direction, independent of location).
- The **column matrix** gives us a precise numerical address to calculate with.

In short, a **vector** is a directed displacement represented as a single column of numbers, whereas a **matrix** is a full grid of numbers that acts as a transformation machine. When a 2×2 matrix A multiplies a vector $\mathbf{x}$, it takes the input vector and transforms it into a new output vector $A\mathbf{x}$.

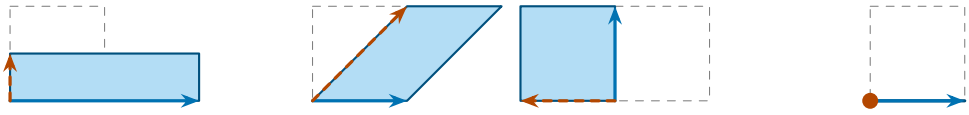
A 2×2 matrix $A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$ is a transformation machine that takes any vector $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ in the plane and outputs a new vector $A\mathbf{x}$. By expressing $\mathbf{x}$ in terms of the standard basis vectors $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$, we see how matrix multiplication is just a **linear combination of the matrix columns**:

$$A\mathbf{x} = A(x\mathbf{i} + y\mathbf{j}) = x(A\mathbf{i}) + y(A\mathbf{j}) = x\begin{bmatrix} a \\ b \end{bmatrix} + y\begin{bmatrix} c \\ d \end{bmatrix}.$$

This means the entire transformation is completely determined by where A sends $\mathbf{i}$ and $\mathbf{j}$:

$$A\mathbf{i} = \text{first column of } A, \qquad A\mathbf{j} = \text{second column of } A.$$

Key Geometric Rule: Linear transformations keep the origin $(0, 0)$ fixed and, when A is invertible, keep grid lines parallel and evenly spaced (a singular matrix collapses the plane onto a line or a point; see panel (d) of the figure below). The original unit square bounded by $\mathbf{i}$ and $\mathbf{j}$ transforms into the parallelogram spanned by $A\mathbf{i}$ and $A\mathbf{j}$, which is flat if A is singular. Figure 1.6 shows what four fundamental 2×2 matrices do to the unit square.



(a) Stretch / Squeeze $\begin{bmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ (b) Shear $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ (c) Rotation (90°) $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (d) Projection $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Figure 1.6: Four 2×2 matrices acting on the unit square (grey dashed outline). The solid blue arrow is the transformed basis vector $A\mathbf{i}$ (first column); the dashed orange arrow is $A\mathbf{j}$ (second column). **(a) Stretch:** scales the x-axis by 2 and squeezes the y-axis by $\frac{1}{2}$ (area is preserved since $\det A = 1$). **(b) Shear:** keeps $\mathbf{i}$ fixed while slanting $\mathbf{j}$ to $(1, 1)$, sliding horizontal lines proportionally to their height. **(c) Rotation:** turns the plane 90° counterclockwise ($\mathbf{i} \to (0, 1)$ and $\mathbf{j} \to (-1, 0)$). **(d) Projection:** collapses the 2D plane onto the 1D x-axis; because $\mathbf{j}$ maps to $(0, 0)$ (orange dot), the matrix has determinant 0, meaning information is destroyed and the action cannot be inverted.

1.3 Real Numbers vs. Matrix Operations

A **zero matrix** (or *null matrix*), denoted O (or simply 0), is a matrix of any size $m \times n$ where every entry is equal to zero. A **square matrix** is a matrix that has an equal number of rows and columns ($m = n$). An $n \times n$ square matrix is said to be of order n.

The **Kronecker delta** is defined as

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j, \end{cases} \qquad i, j \in \{1, 2, \ldots, n\}. \tag{1.1}$$

Using the Kronecker delta, the **identity matrix of order** n, denoted I_n, is the $n \times n$ matrix whose (i, j)-entry equals δ_{ij}:

$$I_n = \begin{bmatrix} \delta_{11} & \delta_{12} & \cdots & \delta_{1n} \\ \delta_{21} & \delta_{22} & \cdots & \delta_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \delta_{n1} & \delta_{n2} & \cdots & \delta_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}. \tag{1.2}$$

To deeply understand matrix algebra, it is helpful to compare it to the arithmetic of real numbers ($\mathbb{R}$). While many properties carry over, matrices introduce critical differences—namely the lack of commutativity and the existence of zero divisors.

In the realm of matrices, the real numbers 0 and 1 are replaced by the **Zero Matrix** (O) and the **Identity Matrix** (I), respectively.

Zero Matrix ($O_{3\times3}$)

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Acts like the number 0.
$A + O = A$

Identity Matrix (I_3)

$$\begin{bmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} \end{bmatrix}$$

Acts like the number 1.
$AI = IA = A$

Table 1.1: Comparison of Real Number and Matrix Algebra Properties

Property	Real Numbers ($\mathbb{R}$)	Matrices ($\mathbb{R}^{n\times n}$)
Additive Identity (Zero)	$a + 0 = 0 + a = a$	$A + O = O + A = A$
Multiplicative Identity (One)	$a \cdot 1 = 1 \cdot a = a$	$AI = IA = A$
Commutativity	$ab = ba$	$AB \neq BA$ (Generally)
Multiplicative Inverse (Division)	$a \cdot a^{-1} = 1$ $(a \neq 0)$	$AA^{-1} = I$ (if $\det(A) \neq 0$)
Zero Divisors	$ab = 0 \implies a = 0$ or $b = 0$	$AB = O \nRightarrow A = O$ or $B = O$

1.3.1 Algebraic Properties

For matrices A, B, C of compatible sizes and scalars k, l:

1. **Commutativity of addition:** $A + B = B + A$.
2. **Associativity:** $(A + B) + C = A + (B + C)$ and $(AB)C = A(BC)$.
3. **Distributivity:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
4. **Scalars:** $k(A + B) = kA + kB$, $(k + l)A = kA + lA$, $k(AB) = (kA)B = A(kB)$.
5. **Identities:** $A + O = A$, and $I_m A = AI_n = A$ for an $m \times n$ matrix A.
6. **Transpose of a product:** $(AB)^T = B^T A^T$ (the order reverses).
7. **Non-commutativity:** in general $AB \neq BA$.

For a *square* matrix we write $A^k = AA \cdots A$ (k factors) and $A^0 = I$.

Critical differences from scalar algebra. Each "rule" below is true for numbers and **false** for matrices.

1. **Zero divisors:** $AB = O$ does **not** imply $A = O$ or $B = O$.
   For instance, with $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ we get $AB = O$, but $BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq O$.
2. **No cancellation:** $AB = AC$ does **not** imply $B = C$ (unless A is invertible); see Problem 2.
3. **Binomial formula:** $(A + B)^2 = A^2 + AB + BA + B^2$, which equals $A^2 + 2AB + B^2$ *only if* $AB = BA$. Likewise $(A + B)(A - B) = A^2 - AB + BA - B^2 \neq A^2 - B^2$ in general.
4. **Powers of a product:** $(AB)^2 = ABAB$ is generally different from $A^2B^2 = AABB$.

Intuition. **The product is "do one, then the other".** $(AB)\mathbf{x} = A(B\mathbf{x})$: first apply B, then apply A. So the **order of the factors is the reverse of the order in time**. Associativity $(AB)C = A(BC)$ then says nothing more than "applying three machines in a row does not depend on how you group them". Order, however, matters a great deal; see Figure 1.7.

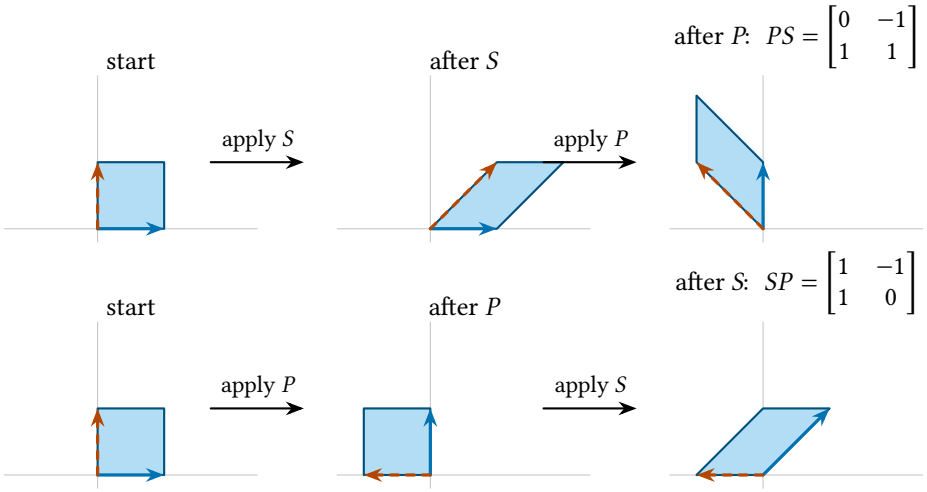


Figure 1.7: $PS \neq SP$. Here $S = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is a shear and $P = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ is a quarter-turn rotation. *Top:* shear first, then rotate. *Bottom:* rotate first, then shear. The two final shapes are different (compare where the blue arrow ends), so the two products are different matrices.

Proof of property 6. The (i,k) entry of $(AB)^T$ is the (k,i) entry of AB, namely $\sum_j a_{kj}b_{ji}$. The (i,k) entry of B^TA^T is $\sum_j (B^T)_{ij}(A^T)_{jk} = \sum_j b_{ji}a_{kj}$. These sums are equal. □

Quick Check 2

1. A is 3×2 and B is 2×4. Is AB defined? Is BA? Give the size of each product that exists.
2. Why does the formula $(AB)^T = A^TB^T$ fail in general? (Test the sizes with A of size 2×3 and B of size 3×4.)
3. In Figure 1.7, which of PS and SP sends $\mathbf{i}$ to $(1,1)$?

1.4 Special Types of Square Matrices

For a square matrix $A = [a_{ij}]_{n\times n}$, the entries $a_{11}, a_{22}, \ldots, a_{nn}$ form the **main diagonal**. Many matrices that appear in practice have a recognizable pattern of zeros, and that pattern makes calculations faster and reveals physical structure (Figure 1.8).

- **Diagonal matrix**: $a_{ij} = 0$ for all $i \neq j$.
- **Identity matrix I_n**: the diagonal matrix with 1s on the main diagonal. For *any* $m \times n$ matrix A we have $I_mA = AI_n = A$.
- **Upper triangular matrix**: $a_{ij} = 0$ for all $i > j$ (everything *below* the main diagonal is zero).
- **Lower triangular matrix**: $a_{ij} = 0$ for all $i < j$ (everything *above* the main diagonal is zero).
- **Symmetric matrix**: $A^T = A$, that is, $a_{ij} = a_{ji}$ (a mirror image across the diagonal).
- **Skew-symmetric matrix**: $A^T = -A$, that is, $a_{ij} = -a_{ji}$. This forces $a_{ii} = 0$.

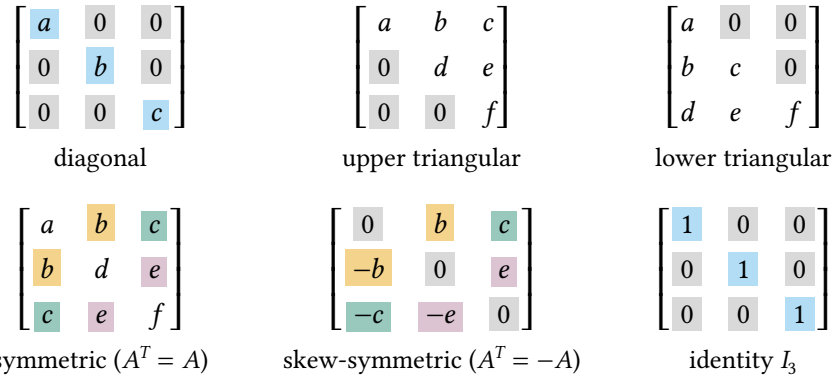


Figure 1.8: Zero patterns of special 3×3 matrices. Grey cells are forced zeros; in the diagonal and identity matrices the blue cells mark the main diagonal. In the symmetric and skew-symmetric matrices, entries with the same color mirror each other across the diagonal (equal in the symmetric case, opposite in sign in the skew case).

Quick Check 3

1. Is a matrix that is both upper and lower triangular necessarily diagonal?
2. Why must the diagonal of a skew-symmetric matrix be zero?

Theorem 1.1: Symmetric plus skew-symmetric

Every square matrix A can be written in exactly one way as the sum of a symmetric and a skew-symmetric matrix:

$$A = \underbrace{\tfrac{1}{2}\left(A + A^T\right)}_{\text{symmetric}} + \underbrace{\tfrac{1}{2}\left(A - A^T\right)}_{\text{skew-symmetric}} .$$

Three facts are used constantly. The first follows from the transpose rules of Section 1.3.1; the other two follow directly from the definition of the product:

1. For *any* matrix A (even a non-square one), A^TA is symmetric, because $(A^TA)^T = A^T(A^T)^T = A^TA$.
2. Powers of a diagonal matrix are computed entry by entry: $\operatorname{diag}(d_1, \dots, d_n)^k = \operatorname{diag}(d_1^k, \dots, d_n^k)$.
3. The product of two upper (or two lower) triangular matrices is again upper (lower) triangular.

1.5 Applications in Physics and Engineering

Matrix algebra is the universal working language across science, engineering, and modern computing. Beyond physical engineering—such as analyzing structural frameworks, electrical circuits, mechanical vibrations, and robotic kinematics—matrices drive digital imagery, economic modeling, quantum physics, and artificial intelligence. Regardless of the discipline, the governing recipe remains the same:

1. **Formulate Local Laws:** Write down the fundamental physical or mathematical relationship governing each individual component—be it Kirchhoff's laws at a circuit node, Newton's second law for a mass in a vibrating system, or pixel transformations in computer graphics.
2. **Assemble the Global System:** Collect all these local equations into a single, unified matrix equation, typically of the form $A\mathbf{x} = \mathbf{b}$ or $\mathbf{x}_{k+1} = A\mathbf{x}_k$.
3. **Deploy Matrix Algebra:** Let the structural tools of linear algebra—determinants, matrix inverses, eigenvalues, and decompositions—solve, transform, and analyze the entire interconnected system at once.

1.5.1 Statics: Forces in a Truss

A **truss** is a structure of straight bars joined at pins. Every bar carries a force only along its own axis: **tension** (pulling on the joints, positive) or **compression** (pushing, negative).

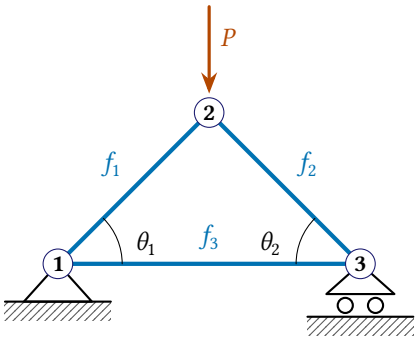
Physics & Engineering: Equilibrium of a Truss

Physical law. A joint at rest has no net force: $\sum F_x = 0$ and $\sum F_y = 0$.
For the triangular truss of Figure 1.9, let f_1, f_2, f_3 be the bar forces (tension > 0) and P the downward load at joint 2. The roller at joint 3 cannot push horizontally. Joint 2 gives two equations and the horizontal balance at joint 3 gives a third:

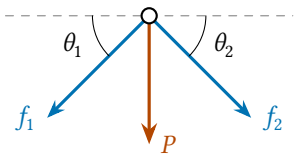
$$\begin{aligned} -f_1\cos\theta_1 + f_2\cos\theta_2 &= 0 && \text{(joint 2, horizontal)}\\ -f_1\sin\theta_1 - f_2\sin\theta_2 - P &= 0 && \text{(joint 2, vertical)}\\ -f_2\cos\theta_2 - f_3 &= 0 && \text{(joint 3, horizontal)} \end{aligned}$$

In matrix form $E\mathbf{f} = \mathbf{p}$:

$$\underbrace{\begin{bmatrix} -\cos\theta_1 & \cos\theta_2 & 0 \\ \sin\theta_1 & \sin\theta_2 & 0 \\ 0 & \cos\theta_2 & 1 \end{bmatrix}}_{E\ \text{(equilibrium matrix)}} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -P \\ 0 \end{bmatrix}.$$



(a) Pin at joint 1, roller at joint 3, load P at joint 2.



(b) Forces on the pin at joint 2 (bars drawn in tension).

Figure 1.9: A triangular truss and the free-body diagram of its top joint. The bar forces are drawn as if the bars were in tension; all three arrows at joint 2 point *downwards*, so the vertical balance can only hold if f_1 and f_2 come out *negative*, that is, if the two upper bars are in compression.

Example: A Symmetric Truss

Take $\theta_1 = \theta_2 = \theta = 45^\circ$ (Figure 1.10), so $\cos\theta = \sin\theta = \frac{1}{\sqrt{2}}$. The first equation gives $f_1 = f_2$. The second gives $2\sin\theta\, f_1 = -P$, so

$$f_1 = f_2 = -\frac{P}{2\sin\theta} = -\frac{P}{\sqrt{2}} \approx -0.71\,P \quad \text{(compression)},$$

$$f_3 = -\cos\theta\, f_2 = \frac{P}{2} \quad \text{(tension)}.$$

The support reactions are $P/2$ each (by symmetry and the vertical balance of joint 3: $R_3 = -f_2\sin\theta = P/2$).

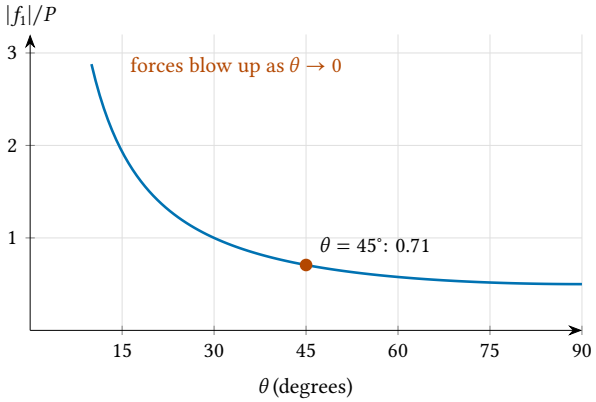


Figure 1.10: Force in each upper bar of the symmetric truss, $|f_1|/P = 1/(2\sin\theta)$. A flat truss (small θ) must carry enormous forces.

Intuition. Why do flat trusses and shallow cables fail so easily? The determinant of E equals $-\sin(\theta_1+\theta_2)$, and for the symmetric truss $\theta_1 = \theta_2 = \theta$ it is $-\sin 2\theta$. As $\theta \to 0$ this tends to 0: the matrix becomes (nearly) *singular*, and the solution of $E\mathbf{f} = \mathbf{p}$ blows up, exactly as in Figure 1.10. (Determinants and singular matrices are treated in the following chapters; here it is enough to know that "determinant close to 0" means "small changes in the data cause huge changes in the answer".)

Common mistake. The relation $F = kx$ (Hooke's law) is *not* the equilibrium equation of a truss joint. Equilibrium involves only forces and angles, and it gives $E\mathbf{f} = \mathbf{p}$. Hooke's law enters later, when we ask how far the joints *move*; see Section 1.5.3.

1.5.2 Electrical Engineering: Mesh-Current Analysis

Physics & Engineering: Mesh-Current Method

Two laws. *Ohm's law:* the voltage drop across a resistor is $V = RI$. *Kirchhoff's voltage law (KVL):* going once around any closed loop, the voltages that push the current equal the voltage drops. In the **mesh-current method** we let a current I_k circulate (say clockwise) around each mesh, apply KVL to each mesh, and obtain one linear equation per mesh. For three meshes in a row, with a source V_1 in the first and V_2 in the last,

$$\begin{bmatrix} R_{11} & -R_{12} & 0 \\ -R_{12} & R_{22} & -R_{23} \\ 0 & -R_{23} & R_{33} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ 0 \\ -V_2 \end{bmatrix}.$$

Here R_{kk} is the total resistance around mesh k, and $R_{jk} = R_{kj}$ is the resistance *shared* by meshes j and k. It appears with a minus sign because, with all currents clockwise, neighboring currents run in opposite directions through the shared resistor. A source enters the right-hand side with $+$ if it pushes the current in the clockwise direction and with $-$ if it opposes it.

Example: Two Meshes

KVL around mesh 1 gives $V_1 = R_1 I_1 + R_3(I_1 - I_2)$, and around mesh 2 gives $-V_2 = R_2 I_2 + R_3(I_2 - I_1)$, see Figure 1.11. In matrix form

$$\begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ -V_2 \end{bmatrix}.$$

For $R_1 = R_2 = 1\,\Omega$, $R_3 = 2\,\Omega$, $V_1 = 10\,\mathrm{V}$, $V_2 = 5\,\mathrm{V}$:

$$\begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ -5 \end{bmatrix} \Longrightarrow \quad I_1 = 4\,\mathrm{A}, \quad I_2 = 1\,\mathrm{A}.$$

(From the first equation $I_1 = (10 + 2I_2)/3$; substituting into the second gives $\frac{5}{3} I_2 = \frac{5}{3}$.)
The current through R_3 is $I_1 - I_2 = 3\,\mathrm{A}$, downward. Check with the outer loop:

$$V_1 - V_2 = 5 = R_1 I_1 + R_2 I_2 = 4 + 1.$$

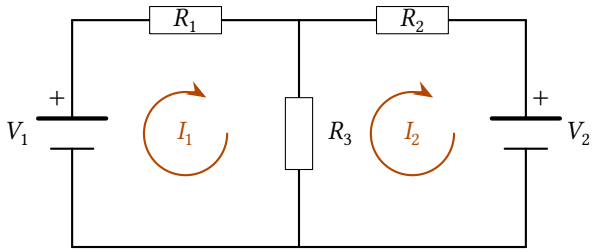


Figure 1.11: A two-mesh circuit. Both mesh currents I_1, I_2 are chosen clockwise. In the shared resistor R_3 the current flows downward for I_1 and upward for I_2, so the net current is $I_1 - I_2$. The longer plate of each battery is its + terminal: V_1 pushes the current clockwise, whereas V_2 pushes against I_2.

1.5.3 Mechanics: Coupled Springs

Physics & Engineering: Spring–Mass Chain

Physical law. Hooke's law: a spring of stiffness k stretched by Δ pulls back with force $k\Delta$. Let two masses on a frictionless line be joined to each other and to two walls by springs k_1, k_2, k_3 (Figure 1.12), pushed by external forces F_1, F_2 so that they move by x_1, x_2. The middle spring stretches by $x_2 - x_1$. In equilibrium each mass has zero net force:

$$\begin{aligned} -k_1 x_1 + k_2(x_2 - x_1) + F_1 &= 0, \\ -k_2(x_2 - x_1) - k_3 x_2 + F_2 &= 0, \end{aligned}$$

that is, $K\mathbf{x} = \mathbf{F}$ with the **stiffness matrix**

$$\begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}.$$

Intuition. The stiffness matrix K is **symmetric**: the external force needed at mass 1 per unit displacement of mass 2 equals the external force needed at mass 2 per unit displacement of mass 1 (both equal $-k_2$). It is also **tridiagonal**: only neighboring masses are connected by a spring, so all other entries are zero. Real structures meshed into thousands of nodes (finite elements) have the same structure, which computers exploit. When the masses move in time, Newton's law reads $M\ddot{\mathbf{x}} + K\mathbf{x} = \mathbf{F}(t)$ with $M = \operatorname{diag}(m_1, m_2)$.

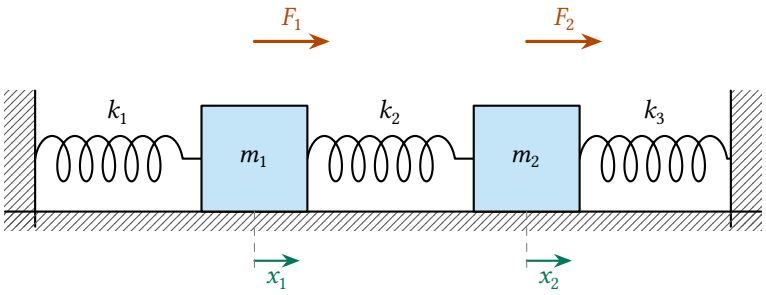


Figure 1.12: Two masses and three springs between two fixed walls. F_1, F_2 are applied forces; x_1, x_2 are the displacements from the rest positions (dashed guides).

Example: Three Equal Springs

Let $k_1 = k_2 = k_3 = k$, $F_1 = 0$ and $F_2 = F$. Then

$$k\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ F \end{bmatrix}.$$

The first row gives $x_2 = 2x_1$; the second then gives $3kx_1 = F$. Hence $x_1 = \dfrac{F}{3k}$ and $x_2 = \dfrac{2F}{3k}$. For $k = 100\,\text{N/m}$ and $F = 30\,\text{N}$: $x_1 = 0.1\,\text{m}$ and $x_2 = 0.2\,\text{m}$.

1.5.4 Rotations and Robot Arms

Physics & Engineering: Rotation Matrices

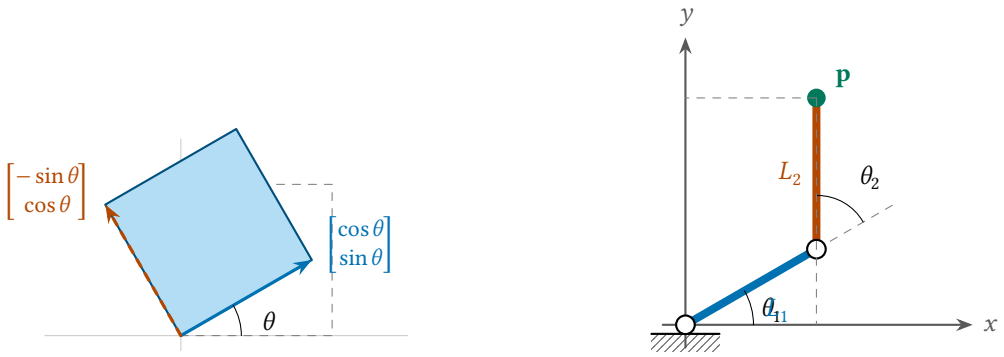
Rotating the plane counterclockwise by the angle θ about the origin is the transformation

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

Its columns are the images of $\mathbf{i}$ and $\mathbf{j}$ (Figure 1.13a). Two key facts:

- **Composition:** $R(\theta)R(\varphi) = R(\theta + \varphi)$. Turning by φ and then by θ is turning by $\theta + \varphi$. (Multiplying out the matrices gives the angle-addition formulas of trigonometry; see Conceptual Problem 10.)
- **Length preservation:** $R(\theta)^T R(\theta) = \begin{bmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \sin^2\theta + \cos^2\theta \end{bmatrix} = I.$
  Hence $\|R\mathbf{v}\|^2 = (R\mathbf{v})^T(R\mathbf{v}) = \mathbf{v}^T R^T R\,\mathbf{v} = \|\mathbf{v}\|^2$.

Matrices with $Q^TQ = I$ are called **orthogonal**; they describe rigid motions that do not stretch anything.



(a) $R(\theta)$ for $\theta = 30^\circ$. **(b)** A two-link arm ($L_1 = L_2 = 1$ drawn at scale 2).

Figure 1.13: (a) Rotation by θ: the first column of $R(\theta)$ (blue) is the new position of $\mathbf{i}$, the second (orange, dashed) the new position of $\mathbf{j}$. (b) A planar robot arm with joint angles θ_1 (measured from the x-axis) and θ_2 (measured from the extension of link 1).

Example: Where Is the Gripper?

Link 1 points at angle θ_1 and link 2 points at angle $\theta_1 + \theta_2$. Using $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$, the tip $\mathbf{p}$ is

$$\mathbf{p} = R(\theta_1)\left(\begin{bmatrix}L_1\\0\end{bmatrix} + R(\theta_2)\begin{bmatrix}L_2\\0\end{bmatrix}\right) = \begin{bmatrix}L_1\cos\theta_1 + L_2\cos(\theta_1+\theta_2)\\ L_1\sin\theta_1 + L_2\sin(\theta_1+\theta_2)\end{bmatrix}.$$

For $L_1 = L_2 = 1$, $\theta_1 = 30^\circ$, $\theta_2 = 60^\circ$: $\mathbf{p} = \big(\cos 30^\circ + \cos 90^\circ,\ \sin 30^\circ + \sin 90^\circ\big) = \big(\tfrac{\sqrt{3}}{2},\ \tfrac{3}{2}\big)$.

Key Formulas of This Chapter

- **Product:** $(m\times n)(n\times p) = (m\times p)$; $c_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk}$; $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$.
- **Transpose:** $(A^T)^T = A$, $(A+B)^T = A^T + B^T$, $(AB)^T = B^TA^T$.
- **Non-commutativity:** $AB \neq BA$ in general, so $(A+B)^2 = A^2 + AB + BA + B^2$.
- **Symmetric/skew split:** $A = \tfrac{1}{2}(A + A^T) + \tfrac{1}{2}(A - A^T)$; A^TA is always symmetric.
- **Rotation:** $R(\theta)R(\varphi) = R(\theta+\varphi)$ and $R(\theta)^TR(\theta) = I$.

1.6 Worked Problems

Problem 1: Matrix Powers

Given $A = \begin{bmatrix}1 & 2\\ 0 & 1\end{bmatrix}$, compute A^2, A^3, and deduce a general formula for A^n, $n \in \mathbb{N}$.

Solution.

$$A^2 = \begin{bmatrix}1&2\\0&1\end{bmatrix}\begin{bmatrix}1&2\\0&1\end{bmatrix} = \begin{bmatrix}1&4\\0&1\end{bmatrix}, \qquad A^3 = A^2A = \begin{bmatrix}1&4\\0&1\end{bmatrix}\begin{bmatrix}1&2\\0&1\end{bmatrix} = \begin{bmatrix}1&6\\0&1\end{bmatrix}.$$

The pattern suggests $A^n = \begin{bmatrix}1 & 2n\\ 0 & 1\end{bmatrix}$. *Proof by induction.* For $n = 1$ this is A. If it holds for n, then

$$A^{n+1} = A^nA = \begin{bmatrix}1&2n\\0&1\end{bmatrix}\begin{bmatrix}1&2\\0&1\end{bmatrix} = \begin{bmatrix}1&2n+2\\0&1\end{bmatrix} = \begin{bmatrix}1&2(n+1)\\0&1\end{bmatrix}. \qquad \square$$

Geometric meaning. A is a shear: it slides the top edge of the unit square two units to the right. Applying it n times slides it $2n$ units (Figure 1.14).

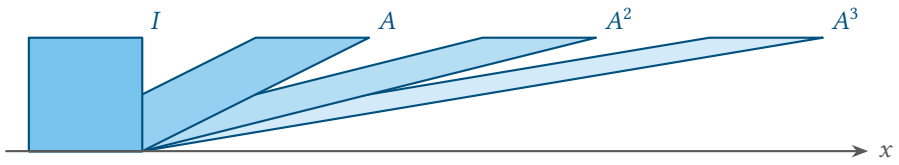


Figure 1.14: The unit square and its images under I, A, A^2, A^3 for $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$. Each application of A shifts the top edge two more units to the right, matching $A^n = \begin{bmatrix} 1 & 2n \\ 0 & 1 \end{bmatrix}$.

Problem 2: Failure of Cancellation

Find a nonzero matrix A and two different matrices $B \neq C$ such that $AB = AC$.

Solution. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} = AC,$$

although $B \neq C$. *Why it happens:* $A\mathbf{x} = (x_1, 0)$ discards the second coordinate (it is the projection of Figure 1.6d), so it cannot "see" that the second rows of B and C differ. Equivalently $A(B - C) = O$ with $B - C \neq O$.

Problem 3: Order Matters

For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ compute AB and BA.

Solution.

$$AB = \begin{bmatrix} 1\cdot 1 + 1\cdot 1 & 1\cdot 0 + 1\cdot 1 \\ 0\cdot 1 + 1\cdot 1 & 0\cdot 0 + 1\cdot 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

So $AB \neq BA$, even though both matrices are 2×2 with entries 0 and 1. Geometrically, A shears to the right and B shears upward, and the two shears do not undo or exchange each other.

Problem 4: Symmetric and Skew-Symmetric Parts

Decompose $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$ as $A = S + K$ with $S^T = S$ and $K^T = -K$.

Solution. By Theorem 1.1, $S = \frac{1}{2}(A + A^T)$ and $K = \frac{1}{2}(A - A^T)$. Here $A^T = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, so

$$S = \tfrac{1}{2}\begin{bmatrix} 2 & 6 \\ 6 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 3 & 3 \end{bmatrix}, \qquad K = \tfrac{1}{2}\begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Indeed $S + K = A$, $S^T = S$ and $K^T = -K$.

Problem 5: AA^T versus A^TA

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ compute AA^T and A^TA. What are their sizes? What do they have in common?

Solution. A is 2×3, so AA^T is 2×2 and A^TA is 3×3:

$$AA^T = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}, \qquad A^TA = \begin{bmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{bmatrix}.$$

Both are symmetric (Section 1.4) and both have trace 91: $14 + 77 = 17 + 29 + 45$. This is no accident: $\operatorname{tr}(AA^T) = \operatorname{tr}(A^TA)$ by Conceptual Problem 4.

1.7 Conceptual & Computational Exercises

1.7.1 Conceptual Problems

Try these *before* touching a calculator: most can be answered by thinking about sizes, symmetry or geometry.

1. **Commutativity analysis.** Find all 2×2 matrices $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ that commute with $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, that is, $XA = AX$. What structure must X have?
2. **Symmetric product property.** If A and B are symmetric $n \times n$ matrices, prove that AB is symmetric **if and only if** A and B commute ($AB = BA$).
3. **Nilpotent matrices.** A matrix N is *nilpotent* if $N^k = O$ for some positive integer k.
   (a) Show that $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is nilpotent.
   (b) Show that $N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ satisfies $N^2 \neq O$ but $N^3 = O$.
   (c) Verify $(I - N)(I + N + N^2) = I$. (It is the matrix version of the geometric series $1/(1 - x) = 1 + x + x^2 + \cdots$.)
4. **Trace.** The trace $\operatorname{tr}(A)$ of a square matrix is the sum of its diagonal entries. Prove that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for any $m \times n$ matrix A and $n \times m$ matrix B. (Note that AB and BA have different sizes!)
5. **True or false? Justify each answer with a proof or a counterexample.**
   (a) If A and B both have size $m \times n$, then AB is defined.
   (b) If AB is defined, then BA is defined.
   (c) If $A^2 = O$, then $A = O$.
   (d) $(A + B)(A - B) = A^2 - B^2$ for all square A, B of the same size.
   (e) If A is symmetric, then A^2 is symmetric.
   (f) A diagonal matrix commutes with every square matrix of the same size.
6. **Size detective.** Let A be 3×2, B be 2×4 and C be 4×3. Which of AB, BA, ABC, CAB, ACB and $(AB)^T$ are defined? Give the size of each one that is.
7. **Geometry of two matrices.**
   (a) What does $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ do to the point (x, y)? Predict P^2 without multiplying, then check.
   (b) What does $\Pi = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ do to (x, y)? Use your answer to explain, in one sentence, why $\Pi B = \Pi C$ can hold with $B \neq C$.

8. **Spot the error.** A student writes $(AB)^T = A^T B^T$ "because transposing is just flipping". Use a 2×3 matrix A and a 3×4 matrix B to show that this cannot be right, and state the correct rule.
9. **Reading the physics.**
   (a) In the truss equation $E\mathbf{f} = \mathbf{p}$, why is the third column of E equal to $(0,0,1)^T$?
   (b) Why is the mesh-resistance matrix symmetric?
   (c) A third mass and a fourth spring k_4 are added to the spring chain. Which entries of the new 3×3 stiffness matrix are nonzero?
10. **Rotations compose.** Without computing, what must $R(\theta)R(\varphi)$ be? Now multiply the two matrices and compare entries to derive the formulas for $\cos(\theta+\varphi)$ and $\sin(\theta+\varphi)$.

1.7.2 Computational Problems

1. Given $A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ -1 & 4 \end{bmatrix}$, evaluate $(A+B)^2$ and compare it with $A^2 + 2AB + B^2$. Explain why they differ, and find the matrix by which they differ.
2. A robot joint rotates its coordinate frame by $M = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. Compute M^2 and $M^T M$. Interpret both results geometrically: what rotation is M^2, and what does $M^T M = I$ say about lengths?
3. For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ compute AB and BA, and verify $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.
4. **Truss.** For the truss of Figure 1.9 with $\theta_1 = 30^\circ$, $\theta_2 = 60^\circ$ and $P = 10\,\text{kN}$, solve $E\mathbf{f} = \mathbf{p}$. State which bars are in tension and which in compression.
5. **Circuit.** In the circuit of Figure 1.11 let $R_1 = 2\,\Omega$, $R_2 = 3\,\Omega$, $R_3 = 6\,\Omega$, $V_1 = 12\,\text{V}$ and $V_2 = 6\,\text{V}$. Write the mesh equations in matrix form and find I_1, I_2 and the current through R_3.
6. **Springs.** In Figure 1.12 let $k_1 = k_3 = 2k$, $k_2 = k$, $F_1 = F$ and $F_2 = 0$. Find x_1 and x_2. Compare with the example in the text and comment on the result.
7. **Robot arm.** For $L_1 = 2$, $L_2 = 1$, $\theta_1 = 60^\circ$ and $\theta_2 = -30^\circ$, find the tip position $\mathbf{p}$ and sketch the arm.

2

Determinants and Inverses

Learning Objectives

After studying this chapter you should be able to:

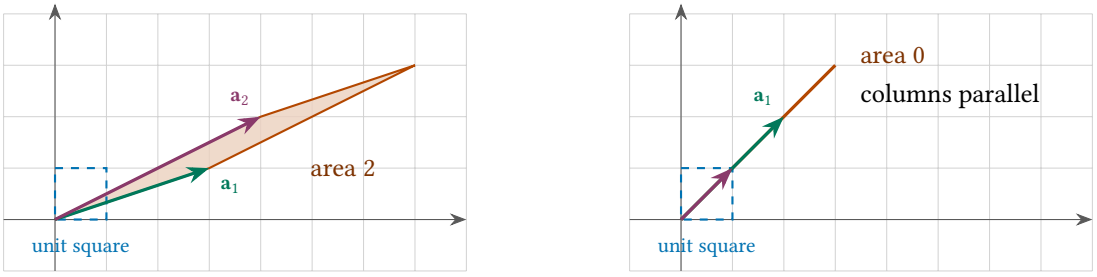
- explain the determinant as a signed area (or volume) and connect $\det A = 0$ with a collapsing transformation;
- compute determinants of 2×2 and 3×3 matrices, and of larger matrices by Laplace expansion and by row reduction;
- use the properties of the determinant (and know what each row operation does to it);
- decide whether a matrix is invertible, and find A^{-1} by the adjugate and by Gauss–Jordan elimination;
- apply determinants to areas and volumes, circuits, structures and geometric transformations.

2.1 Determinants

2.1.1 What the Determinant Measures

Let A be a square matrix. The **determinant**, written $\det(A)$ or $|A|$, is a scalar. Think of A as a transformation of the plane (or of space). It sends the standard basis vectors $\mathbf{i}, \mathbf{j}$ to the *columns* of A, so it sends the unit square to the parallelogram spanned by those columns, see Figure 2.1. Then:

- $|\det A|$ is the **factor by which areas are multiplied** (volumes, in 3D);
- the **sign** of $\det A$ tells whether the orientation is preserved (+) or flipped like a mirror image (−);
- $\det A = 0$ means the parallelogram is **flattened** onto a line (or a point): the transformation destroys information and cannot be undone, so A has no inverse.



(a) $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$, $\det A = 2$ (b) $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$, $\det A = 0$

Figure 2.1: The unit square (blue, dashed) and its image under A. (a) The columns $\mathbf{a}_1 = (3, 1)$ and $\mathbf{a}_2 = (4, 2)$ span a parallelogram of area $|3 \cdot 2 - 4 \cdot 1| = 2$: areas are doubled. (b) The columns $(2, 2)$ and $(1, 1)$ are parallel, so the whole square is squashed onto a line: area 0, and A cannot be inverted.

Common mistake. The sign of the determinant. A negative determinant does *not* mean "negative area". The area (or volume) is $|\det A|$; the sign only records whether the orientation is preserved or flipped. Also, only **square** matrices have a determinant.

In three dimensions the columns of a 3×3 matrix span a **parallelepiped** and $|\det A|$ is its volume, see Figure 2.2.

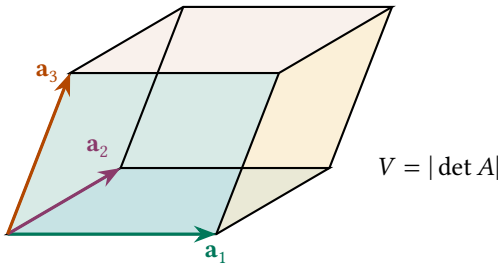


Figure 2.2: The columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ of a 3×3 matrix span a parallelepiped whose volume is $|\det A|$. If the three columns lie in one plane, the box is flat, the volume is 0, and the matrix is singular.

2.1.2 Determinant of a 2×2 Matrix

For a 1×1 matrix $A = [a]$, the determinant is the number itself: $|A| = a$. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Its **determinant** is

$$\det(A) = ad - bc \qquad \text{(main diagonal minus anti-diagonal).}$$

Geometrically, $\det(A)$ is the signed area of the parallelogram **spanned** by the column vectors $\mathbf{v}_1 = \begin{bmatrix} a \\ c \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} b \\ d \end{bmatrix}$, see Figure 2.3.

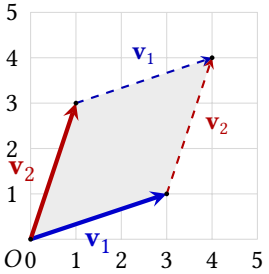


Figure 2.3: The parallelogram spanned by $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. The area of this shaded region equals $|\det([\mathbf{v}_1\ \mathbf{v}_2])| = |(3)(3) - (1)(1)| = 8$.

What does "spanned" mean? The **span** of two vectors is the set of all possible linear combinations you can build using them:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 \quad \text{for all real numbers } c_1, c_2.$$

1. As a Grid/Region: If you start at the origin and walk along $\mathbf{v}_1$ and $\mathbf{v}_2$, the region bounded between 0, $\mathbf{v}_1$, $\mathbf{v}_2$, and their sum $\mathbf{v}_1 + \mathbf{v}_2$ forms a tilted 2D parallelogram. The determinant measures the exact *area* of this shape.

2. Dimensionality (Plane vs. Line):

- If $\mathbf{v}_1$ and $\mathbf{v}_2$ point in different directions, their span fills out the **entire 2D plane** ($\det(A) \neq 0$).
- If $\mathbf{v}_1$ and $\mathbf{v}_2$ lie along the same line (linearly dependent), the parallelogram collapses into a flat line segment with **zero area** ($\det(A) = 0$).

Intuition. Where $ad - bc$ comes from. Draw the parallelogram spanned by (a, c) and (b, d) inside the rectangle of width $a + b$ and height $c + d$ (take $a, b, c, d > 0$ with $ad > bc$). The rectangle has area $(a+b)(c+d) = ac + ad + bc + bd$. Cut away the two triangles of area $\frac{1}{2}ac$, the two triangles of area $\frac{1}{2}bd$ and the two small rectangles of area bc, a total of $ac + bd + 2bc$. What remains is $ad - bc$. The formula is not an arbitrary recipe: it is area bookkeeping.

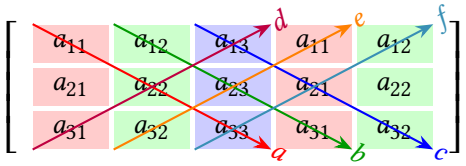
Example: Determinant of a 2×2 Matrix

For $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$: $\det(A) = 3 \cdot 2 - 4 \cdot 1 = 2$ (the area 2 of Figure 2.1a).

Swap the two columns: $A' = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$ has $\det(A') = 4 \cdot 1 - 3 \cdot 2 = -2$. The same parallelogram, but traversed in the opposite orientation: the sign flips.

2.1.3 Determinant of a 3×3 Matrix: Sarrus' Rule (Basket-Weave)

For $A = [a_{ij}]$ of size 3×3, repeat the first two columns to the right of the matrix. Multiply along the three downward diagonals (positive terms a, b, c) and along the three upward diagonals (negative terms d, e, f):

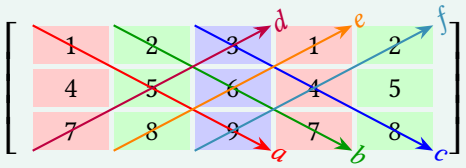


$$\det(A) = \underbrace{a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}}_{\text{downward: } a+b+c} - \underbrace{\left(a_{31}a_{22}a_{13} + a_{32}a_{23}a_{11} + a_{33}a_{21}a_{12}\right)}_{\text{upward: } d+e+f}.$$

Common mistake. Sarrus' rule works only for 3×3 matrices. A 4×4 determinant has $4! = 24$ terms, not the 8 that a similar "diagonal" scheme would give, so the pattern does *not* extend. For larger matrices use Laplace expansion or row reduction.

Example: Sarrus' Rule

Compute $\det(A)$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$.



Downward: $1 \cdot 5 \cdot 9 = 45$, $2 \cdot 6 \cdot 7 = 84$, $3 \cdot 4 \cdot 8 = 96$; sum $= 225$.

Upward: $7 \cdot 5 \cdot 3 = 105$, $8 \cdot 6 \cdot 1 = 48$, $9 \cdot 4 \cdot 2 = 72$; sum $= 225$.

Hence $\det(A) = 225 - 225 = 0$.

2.1.4 Determinant of an $n \times n$ Matrix: Laplace Expansion

Laplace expansion reduces an $n \times n$ determinant to determinants of size $n-1$, and works for every n.

Definition: Minor and Cofactor

Let $A = [a_{ij}]$ be $n \times n$. The **minor** M_{ij} of the entry a_{ij} is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting the i-th row and j-th column of A. The **cofactor** of a_{ij} is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The factor $(-1)^{i+j}$ gives a checkerboard of signs, starting with + in the upper-left corner:

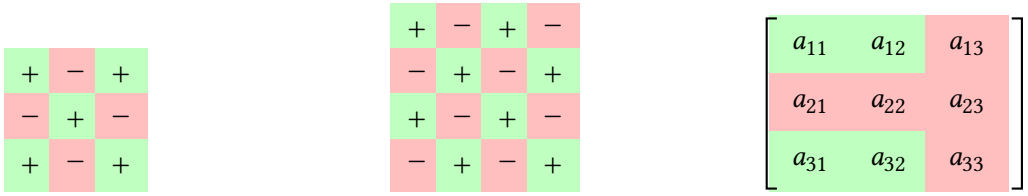


Figure 2.4: Left and middle: the sign $(-1)^{i+j}$ attached to each position. Right: to find the minor M_{23} delete row 2 and column 3 (red); the remaining green entries form the smaller matrix whose determinant is M_{23}.

Example: Minors and Cofactors

For $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$:

C_{11}: delete row 1 and column 1: $M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22}a_{33} - a_{23}a_{32}$, and $C_{11} = (-1)^{1+1} M_{11} = M_{11}$.

C_{23}: delete row 2 and column 3 (Figure 2.4, right): $M_{23} = \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} = a_{11}a_{32} - a_{12}a_{31}$, and $C_{23} = (-1)^{2+3} M_{23} = -M_{23}$.

Theorem 2.1: Laplace expansion

The determinant of an $n \times n$ matrix A can be computed by expanding along **any** row i or **any** column j:

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \quad \text{or} \quad \det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

Intuition. **Choosing the row or column.** Terms with $a_{ij} = 0$ vanish, so expand along the row or column with the **most zeros**: it saves the most minors. Expanding along row 1 of a 3×3 matrix gives

$$\det(A) = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Intuition. **Why the signs form a checkerboard.** The determinant is linear in each row, so row $i = (a_{i1}, \dots, a_{in})$ can be split as $a_{i1}\mathbf{e}_1 + \cdots + a_{in}\mathbf{e}_n$ and each piece treated separately. The piece with the single entry a_{ij} can be moved to the top-left corner using $i-1$ row swaps and $j-1$ column swaps; every swap flips the sign (Property 1), giving $(-1)^{i+j-2} = (-1)^{i+j}$. What is left is the minor. So the checkerboard is just bookkeeping of orientation flips.

Common mistake. **Cofactor expansion.**

- The sign $(-1)^{i+j}$ belongs to the *position*, not to the number sitting there. For $a_{12} = -3$ the term is $a_{12}C_{12} = (-3)(-M_{12}) = +3M_{12}$.
- A *minor* is not a *cofactor*: forgetting $(-1)^{i+j}$ is the most frequent source of wrong answers.
- Use the entries of *one* row (or column) with the cofactors of *that same* row (column). Entries of row i with cofactors of a different row give 0, which can hide a mistake.

Example: Expansion Along the First Row

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$: $\det(A) = 1\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = 1(-3) - 2(-6) + 3(-3) = 0$, in agreement with Sarrus' rule.

Example: A 4×4 Determinant

Find $\det(A)$ for $A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 3 & 5 & 0 & 6 \\ 7 & 8 & 0 & 9 \\ 2 & 1 & 3 & 3 \end{bmatrix}$. Column 3 contains three zeros, so expand along it:

$\det(A) = a_{43}C_{43} = 3\,C_{43}$ with $C_{43} = (-1)^{4+3}M_{43} = -M_{43}$, where

$$M_{43} = \begin{vmatrix} 1 & 2 & 4 \\ 3 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 1(45-48) - 2(27-42) + 4(24-35) = -3 + 30 - 44 = -17.$$

So $C_{43} = 17$ and $\det(A) = 3 \cdot 17 = \mathbf{51}$.

Intuition. **Alien cofactors: why the adjugate formula works.** If you multiply the entries of row i by the cofactors of a *different* row $k \neq i$, the sum is always 0:

$$a_{i1}C_{k1} + a_{i2}C_{k2} + \cdots + a_{in}C_{kn} = 0 \qquad (i \neq k).$$

(It is the expansion of a determinant that has two equal rows.) Together with Laplace expansion ($i = k$ gives $\det A$) this says $A \cdot [C_{kj}]^T = \det(A)\, I$, the key to the inverse formula in Section 2.3.2.

2.1.5 How Expensive Is Cofactor Expansion?

Expanding recursively produces $n!$ products, which explodes. Row reduction (Section 2.2) needs only about $\frac{1}{3}n^3$ multiplications.

n	terms $n!$	row red. $\approx n^3/3$	time for $n!$ at 10^9/s
3	6	9	instant
5	120	42	instant
10	3 628 800	333	4 ms
15	$1.3 \cdot 10^{12}$	1 125	22 min
20	$2.4 \cdot 10^{18}$	2 667	about 77 years

Table 2.1: Cost of computing a determinant. Cofactor expansion is perfect for small matrices or matrices with many zeros, hopeless for large ones; software always uses row reduction (LU decomposition).

2.2 Properties of the Determinant

We write R_i for the i-th row and C_j for the j-th column. The three **elementary row operations** are

$$\begin{aligned} R_i &\leftrightarrow R_j && \text{swap rows } i \text{ and } j, \\ R_i &\to kR_i && \text{multiply row } i \text{ by } k \neq 0, \\ R_i &\to R_i + kR_j && \text{add } k \text{ times row } j \text{ to row } i \ (i \neq j). \end{aligned}$$

1. **Swap.** Interchanging two rows (or two columns) changes the sign of the determinant: $\det \mapsto -\det$.
2. **Scaling.** Multiplying one row (or column) by k multiplies the determinant by k: $\det \mapsto k \det$.
3. **Replacement.** Adding a multiple of one row (column) to another row (column) does **not** change the determinant.
4. **Linearity in a row.** If row i is a sum $\mathbf{r}+\mathbf{s}$, then the determinant is the sum of the two determinants with row i replaced by $\mathbf{r}$ and by $\mathbf{s}$ (other rows unchanged). But $\det(A+B) \neq \det A + \det B$ in general.
5. **Zero row or column.** If a row or a column is zero, then $\det = 0$.
6. **Equal or proportional rows.** If two rows (or columns) are identical, or one is a multiple of another, then $\det = 0$.
7. **Triangular matrices.** For an upper or lower triangular matrix the determinant is the product of the diagonal entries:
$$\det \begin{bmatrix} a_{11} & * & * & \cdots & * \\ 0 & a_{22} & * & \cdots & * \\ 0 & 0 & a_{33} & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix} = a_{11}a_{22}\cdots a_{nn}.$$
8. **Transpose.** $\det(A^T) = \det(A)$. (So every statement about rows is also true for columns.)
9. **Multiplicativity.** $\det(AB) = \det(A)\det(B)$.
10. **Scalar multiple.** For an $n \times n$ matrix A: $\det(kA) = k^n \det(A)$.

Row operation	Effect on det	Geometric meaning
$R_i \leftrightarrow R_j$	$\times(-1)$	orientation reversed (mirror image)
$R_i \to kR_i$	$\times k$	one side of the parallelogram stretched by k
$R_i \to R_i + kR_j$	$\times 1$	a *shear*: base and height unchanged, so area unchanged

Table 2.2: Effect of row operations on the determinant (Properties 1–3).

Common mistake. Row operations.

- Only $R_i \to R_i + kR_j$ leaves det unchanged. The operation $R_i \to kR_i + R_j$ replaces row i by a multiple of itself, so it *multiplies* det by k.
- When you scale a row to simplify, compensate: if you divide a row by k to get smaller numbers, the original determinant is k times the new one.
- Count every swap. Two swaps cancel, three do not.
- For solving systems you may scale rows freely; for determinants you must keep a record of every scaling and swap.

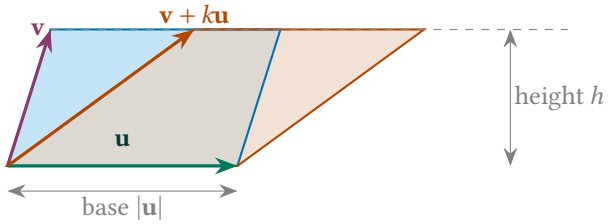


Figure 2.5: Replacement $R_2 \to R_2 + kR_1$ in rows-as-sides language: replacing $\mathbf{v}$ by $\mathbf{v} + k\mathbf{u}$ slides the top edge of the parallelogram sideways. Base and height are unchanged, so the area (the determinant) is unchanged.

Example: Reduction to Triangular Form

Compute $\det(A)$ for $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -2 & 2 \\ 6 & 1 & -1 \end{bmatrix}$ using only replacements (Property 3), which do not change the determinant.

Step 1 ($R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$): $\begin{bmatrix} 2 & 1 & 3 \\ 0 & -4 & -4 \\ 0 & -2 & -10 \end{bmatrix}$.

Step 2 ($R_3 \to R_3 - \frac{1}{2}R_2$): $\begin{bmatrix} 2 & 1 & 3 \\ 0 & -4 & -4 \\ 0 & 0 & -8 \end{bmatrix}$.

The matrix is upper triangular (Property 7), so $\det(A) = (2)(-4)(-8) = \mathbf{64}$.

Intuition. Elimination reshapes the parallelepiped. A replacement is a shear (Figure 2.5): it slides the shape sideways without changing base or height, so the volume is unchanged. A sequence of shears straightens a parallelepiped into a rectangular box whose sides are the diagonal entries; its volume is their product. Swaps flip the orientation (a sign) and scalings stretch one side (a factor). That is the whole algorithm for computing a determinant by elimination.

Intuition. Why $\det(AB) = \det A \det B$ and $\det(kA) = k^n \det A$. AB means "first transform by B, then by A". B multiplies areas by $\det B$ and then A multiplies the result by $\det A$, so the overall factor is the product. For kA each of the n sides of the box is stretched by k: doubling every side of a cube multiplies its volume by $2^3 = 8$, not by 2.

Common mistake. **The determinant is not additive.** $\det(A+B) \neq \det A + \det B$ and $\det(kA) \neq k\det A$. The determinant is linear in *one row at a time*, not in the whole matrix. For $A = B = I_2$: $\det(A+B) = 4$ but $\det A + \det B = 2$.

Example: When a Swap Is Needed

A zero in the pivot position forces a row swap, and every swap flips the sign.

Let $A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 1 & 3 \end{bmatrix}$.

$R_1 \leftrightarrow R_2$ gives $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ (one swap, so det is multiplied by -1); $R_3 \to R_3 - 2R_1$ gives $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & -1 & 3 \end{bmatrix}$;

$R_3 \to R_3 + \frac{1}{2}R_2$ gives $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & \frac{7}{2} \end{bmatrix}$.

The triangular determinant is $1 \cdot 2 \cdot \frac{7}{2} = 7$, so $\det(A) = -7$. (Check by expanding along row 1: $0 - 2(3-0) + 1(1-2) = -7$.)

Example: Multiplicativity and $\det(kA)$

For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ($\det = -2$) and $B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ ($\det = -1$): $AB = \begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix}$ and $\det(AB) = 14 - 12 = 2 = (-2)(-1)$. Also $\det(3A) = \begin{vmatrix} 3 & 6 \\ 9 & 12 \end{vmatrix} = 36 - 54 = -18 = 3^2 \cdot (-2)$, not $3 \cdot (-2)$.

The sum rule fails: $\det(A+B) = \begin{vmatrix} 1 & 3 \\ 4 & 5 \end{vmatrix} = -7 \neq -2 + (-1)$.

2.3 Invertible Matrices

Definition: Invertible Matrix

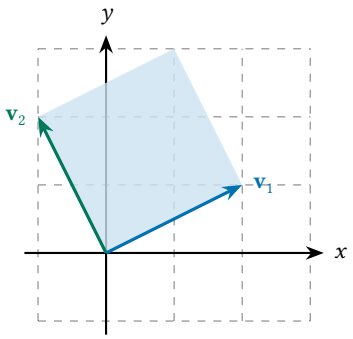
An $n \times n$ matrix A is **invertible** (or **nonsingular**) if there is an $n \times n$ matrix B such that $AB = BA = I_n$, where I_n is the identity matrix. Then B is unique and is called the **inverse** of A, written A^{-1}. A matrix with no inverse is **singular**.

Theorem 2.2: Invertibility criteria

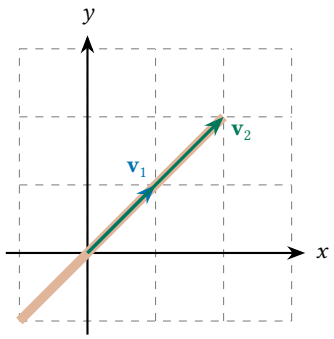
For an $n \times n$ matrix A the following statements are equivalent:

1. A is invertible;
2. $\det(A) \neq 0$;
3. $A\mathbf{x} = \mathbf{0}$ has only the solution $\mathbf{x} = \mathbf{0}$;
4. $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$;
5. A can be reduced to the identity matrix by row operations.

Intuition. Invertible means "nothing is lost". A singular matrix squashes space, so different inputs land on the same output. The projection $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ sends both $(1, 2)$ and $(1, 5)$ to $(1, 0)$; given only the output $(1, 0)$, no inverse can tell which input it came from. An invertible matrix never merges points, so every output has exactly one input. This one picture explains all five criteria: a nonzero $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$ (the inputs that get squashed to 0), no unique solution, no reduction to I, and $\det A = 0$.



$\det(A) \neq 0$ **(Invertible)**
Area is not collapsed.
Transformation can be reversed.

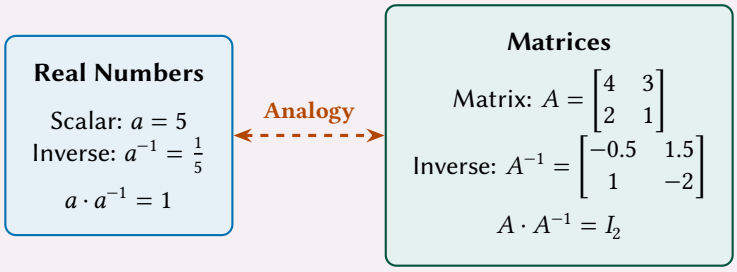


$\det(A) = 0$ **(Singular)**
Area collapses to a line.
Information is lost permanently.

Analogy: Multiplicative Inverse of Real Numbers

Just as the multiplicative inverse of a non-zero real number a is $a^{-1} = 1/a$ where $a \cdot a^{-1} = 1$, the inverse of a square matrix A is A^{-1} where $A \cdot A^{-1} = I_n$.

In real numbers, you cannot divide by zero (0 has no inverse). In linear algebra, a matrix with $\det(A) = 0$ similarly has no inverse.



Common mistake. Judging singularity by the size of $\det A$**.** Only $\det A = 0$ *exactly* means singular. A small determinant proves nothing: changing units changes it. The same 2×2 stiffness matrix has a determinant 10^6 times larger in N/m than in N/mm. How close a matrix is to being singular is measured by its *condition number*, not by the determinant (see Section 2.5).

2.3.1 The Inverse of a 2×2 Matrix

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $\det(A) = ad - bc \neq 0$, $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Memory aid: *swap* the diagonal entries, *negate* the off-diagonal entries, *divide* by the determinant. Since $\det(A^{-1}) = \frac{1}{\det A}$, a transformation that doubles areas is undone by one that halves them.

Example: Inverse of a 2×2 Matrix

Find the inverse of $A = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$.

$\det(A) = 4 \cdot 1 - 3 \cdot 2 = -2 \neq 0$, so A is invertible and

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{3}{2} \\ 1 & -2 \end{bmatrix}.$$

2.3.2 Inverse of an $n \times n$ Matrix: the Adjugate

The **adjugate** (or classical adjoint) of A is the transpose of the cofactor matrix $C = [C_{ij}]$:

$$\operatorname{adj}(A) = C^T, \qquad C_{ij} = (-1)^{i+j} M_{ij}.$$

By Laplace expansion and the alien-cofactor rule, $A \operatorname{adj}(A) = \det(A)\, I$, hence

Theorem 2.3: Adjugate formula

If $\det(A) \neq 0$ then $A^{-1} = \dfrac{1}{\det(A)} \operatorname{adj}(A)$.

Example: Inverse of a 3×3 Matrix by the Adjugate

Compute A^{-1} for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$.

1. Determinant (expand along row 2, which has a zero):

$$\det(A) = 4\,(-1)^{2+2} \begin{vmatrix} 1 & 3 \\ 1 & 6 \end{vmatrix} + 5\,(-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = 4(6-3) - 5(0-2) = 12 + 10 = 22 \neq 0.$$

2. Cofactors:

$$C_{11} = +\begin{vmatrix} 4 & 5 \\ 0 & 6 \end{vmatrix} = 24, \quad C_{12} = -\begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} = 5, \quad C_{13} = +\begin{vmatrix} 0 & 4 \\ 1 & 0 \end{vmatrix} = -4,$$

$$C_{21} = -\begin{vmatrix} 2 & 3 \\ 0 & 6 \end{vmatrix} = -12, \quad C_{22} = +\begin{vmatrix} 1 & 3 \\ 1 & 6 \end{vmatrix} = 3, \quad C_{23} = -\begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = 2,$$

$$C_{31} = +\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = -2, \quad C_{32} = -\begin{vmatrix} 1 & 3 \\ 0 & 5 \end{vmatrix} = -5, \quad C_{33} = +\begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} = 4.$$

3. Adjugate (transpose of the cofactor matrix):

$$C = \begin{bmatrix} 24 & 5 & -4 \\ -12 & 3 & 2 \\ -2 & -5 & 4 \end{bmatrix}, \qquad \operatorname{adj}(A) = C^T = \begin{bmatrix} 24 & -12 & -2 \\ 5 & 3 & -5 \\ -4 & 2 & 4 \end{bmatrix}.$$

4. Inverse:

$$A^{-1} = \frac{1}{22} \begin{bmatrix} 24 & -12 & -2 \\ 5 & 3 & -5 \\ -4 & 2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{12}{11} & -\frac{6}{11} & -\frac{1}{11} \\ \frac{5}{22} & \frac{3}{22} & -\frac{5}{22} \\ -\frac{2}{11} & \frac{1}{11} & \frac{2}{11} \end{bmatrix}.$$

Common mistake. The adjugate.

- The adjugate is the *transpose* of the cofactor matrix. Forgetting the transpose gives $(A^{-1})^T$ instead of A^{-1}.
- Write the checkerboard of signs first (Figure 2.4); one wrong sign, for example in C_{13}, ruins a whole row or column of the answer.
- Always verify with $A\,\mathrm{adj}(A) = \det(A)\,I$: an entry that comes out wrong points to the offending row of A and column of $\mathrm{adj}(A)$.
- The formula is exact and elegant, but needs n^2 determinants of size $n-1$; for numerical work use Gauss–Jordan or an LU decomposition.

2.3.3 Inverse by Gauss–Jordan Elimination

Write A and the identity side by side, $[\,A \mid I\,]$, and apply row operations until the left block becomes I. The *same* operations turn the right block into A^{-1}:

$$[\,A \mid I\,] \xrightarrow{\text{row operations}} [\,I \mid A^{-1}\,].$$

If the left block cannot be reduced to I (a zero row appears), A is singular.

Example: Gauss–Jordan

Find the inverse of $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ ($\det A = 2$).

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & -1 & 0 & 1 \end{array}\right] \xrightarrow{R_3 + R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right] \xrightarrow{R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right] \xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{array}\right].$$

Hence $A^{-1} = \dfrac{1}{2}\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$.

Intuition. Why Gauss–Jordan finds the inverse. Each row operation is multiplication on the left by an elementary matrix E. If operations $E_1, \dots, E_k$ turn A into I, then $E_k \cdots E_1 A = I$, so $E_k \cdots E_1 = A^{-1}$. Applying the very same operations to I gives $E_k \cdots E_1 I = A^{-1}$: the right half of $[\,A \mid I\,]$ is a running record of everything we did to A.

2.3.4 Properties of Invertible Matrices

Let A, B be invertible $n \times n$ matrices and $k \neq 0$ a scalar.

1. **Uniqueness.** A^{-1} is unique. (If B and B' were both inverses: $B = BAB' = B'$.)
2. **Inverse of a product.** AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$: the order *reverses* (put on socks, then shoes; undo shoes first, then socks).
3. **Inverse of the inverse.** $(A^{-1})^{-1} = A$.
4. **Transpose.** A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

5. **Determinant.** $\det(A^{-1}) = \dfrac{1}{\det(A)}$ (from $\det(A)\det(A^{-1}) = \det(I) = 1$).
6. **Scalar multiple.** kA is invertible and $(kA)^{-1} = \frac{1}{k}A^{-1}$.
7. **Powers.** $(A^m)^{-1} = (A^{-1})^m$.
8. **Cancellation.** If $AB = AC$ then $B = C$; if $BA = CA$ then $B = C$ (multiply by A^{-1} on the left, resp. right).
9. **One-sided inverses suffice.** For square matrices, $AB = I$ already implies $BA = I$.

Common mistake. **What is *not* true.**

- $(A + B)^{-1} \neq A^{-1} + B^{-1}$ in general (and $A + B$ may even be singular, e.g. $I + (-I) = 0$).
- $(AB)^{-1} \neq A^{-1}B^{-1}$ in general; the correct order is $B^{-1}A^{-1}$.
- $AB = AC$ does *not* imply $B = C$ if A is singular (for example $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$).

2.3.5 Linear Matrix Equations

A **linear matrix equation** has the form

$$AX + B = C,$$

with A an $n \times n$ matrix, X the unknown $n \times p$ matrix, and B, C given $n \times p$ matrices. If A is invertible, subtract B and multiply *from the left* by A^{-1}:

$$AX = C - B \quad \Longrightarrow \quad X = A^{-1}(C - B),$$

and the solution is unique. For $XA + B = C$ multiply from the *right*: $X = (C - B)A^{-1}$. Matrix multiplication is not commutative, so $A^{-1}(C - B)$ and $(C - B)A^{-1}$ are generally different.

Common mistake. **Order of multiplication.** In $AX = D$ multiply both sides by A^{-1} *on the left*: $X = A^{-1}D$. Writing $X = DA^{-1}$ is wrong in general (it solves $XA = D$). Do not cancel A from the middle of a product, and check that B and C in $AX + B = C$ have the same size as AX.

2.4 Determinants and Inverses in Physics and Engineering

Physics & Engineering

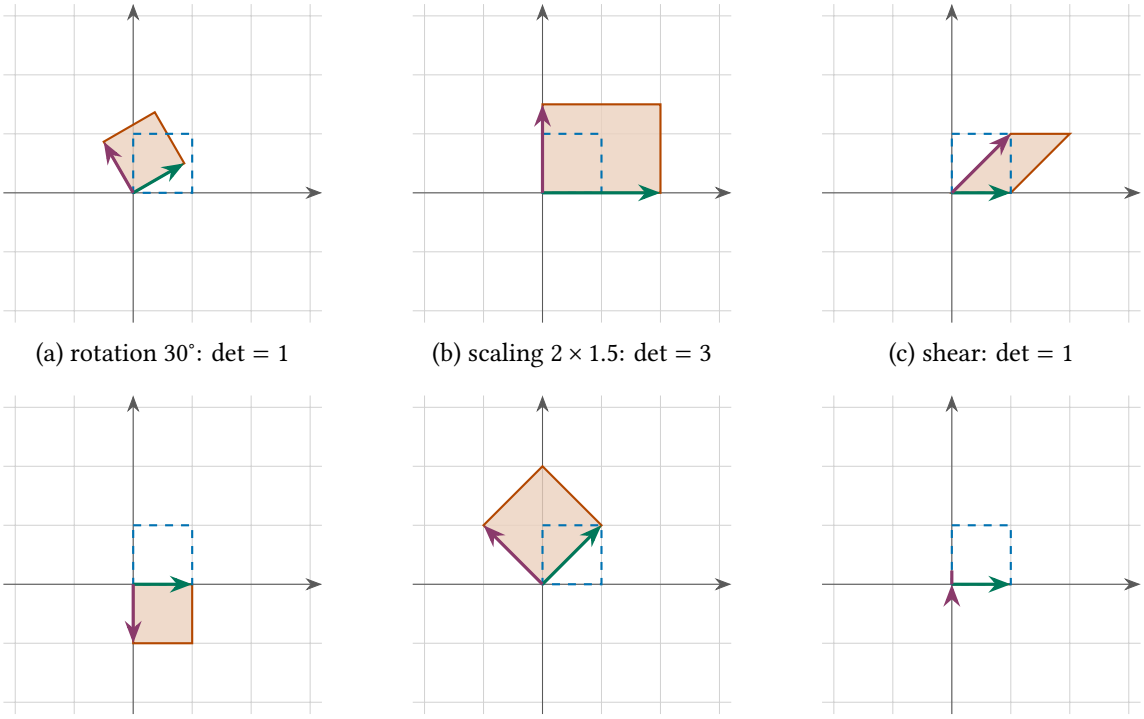
Whenever several quantities are coupled linearly (forces in a structure, currents in a network, coordinates in a robot arm) the model is a matrix equation, and the determinant decides whether the problem is well posed. This section collects the most important uses.

2.4.1 Cryptography: the Hill Cipher

Messages can be encrypted by multiplying blocks of letters (A= 0, … ,Z= 25) by a key matrix modulo 26. Decryption needs the *inverse of the key matrix modulo* 26, which exists only if $\det K$ has no common factor with 26. For $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$: $\det K = 9$ and $\gcd(9, 26) = 1$, so K is a valid key, and $K^{-1} \equiv \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \pmod{26}$. The word HI $= (7, 8)$ is encrypted to $K(7, 8)^T = (45, 54) \equiv (19, 2) =$ TC, and $K^{-1}(19, 2)^T = (319, 398) \equiv (7, 8)$ recovers it.

2.4.2 Geometric Transformations: Graphics, Robotics, Mechanics

Rotations, scalings, reflections and shears of the plane are all given by 2×2 matrices, and their determinants describe what they do to area and orientation (Figure 2.6 and Table 2.3).



(a) rotation 30°: det = 1 (b) scaling 2×1.5: det = 3 (c) shear: det = 1

(d) reflection: det = −1 (e) rotate 45° and scale $\sqrt{2}$: det = 2 (f) projection: det = 0

Figure 2.6: The unit square (blue, dashed) and its image (red) under six transformations; the green and purple arrows are the images of $\mathbf{e}_1$ and $\mathbf{e}_2$, i.e. the columns of the matrix. Rotations and shears preserve area, scalings multiply it, a reflection flips orientation (negative determinant), and a projection flattens the square to a segment (determinant zero, not invertible).

Transformation	Matrix	det	Inverse
rotation by θ	$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$	1	rotation by $-\theta$; $R^{-1} = R^T$
scaling	$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$	$s_x s_y$	scaling by $1/s_x, 1/s_y$
shear	$\begin{bmatrix} 1 & m \\ 0 & 1 \end{bmatrix}$	1	shear by $-m$
reflection in the x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	−1	itself
projection onto the x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	0	**none**

Table 2.3: Basic plane transformations. For a composition the determinants multiply (Property 9): rotating (det 1), scaling by 2×1.5 (det 3), then reflecting (det −1) gives a map with determinant −3: areas triple and the orientation flips.

Robotics & Mechanics: Rotation Matrices

A rotation matrix R of the plane or of space is **orthogonal** ($R^T R = I$) with $\det R = +1$ (a proper rotation, a right-handed frame stays right-handed). Hence $R^{-1} = R^T$: the inverse of a rotation is found by transposing, with no determinant or elimination needed. For the 90° rotation $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $R^{-1} = R^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. A matrix with $\det = -1$ includes a reflection, and would turn a right-handed coordinate system into a left-handed one. In continuum mechanics the determinant of the deformation gradient is the ratio of deformed to original volume: $\det F = 1$ for an incompressible material.

2.4.3 Electric Circuits: Mesh Currents

Kirchhoff's voltage law around each loop of a resistor network gives a linear system $R\mathbf{I} = \mathbf{V}$, with the resistance matrix R and the source voltages $\mathbf{V}$. A unique set of currents exists exactly when $\det R \neq 0$ (which is always true for a network of positive resistors).

Example: Two-Loop Network

Voltage law for loop 1: $R_1 I_1 + R_2(I_1 - I_2) = V$; for loop 2: $R_3 I_2 + R_2(I_2 - I_1) = 0$. With the values of Figure 2.7:

$$\begin{bmatrix} 5 & -3 \\ -3 & 7 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}, \qquad \det R = 35 - 9 = 26 \neq 0.$$

$$I_1 = \frac{35}{13} \approx 2.69 \text{ A}, \quad I_2 = \frac{15}{13} \approx 1.15 \text{ A}.$$

Inverse: $R^{-1} = \frac{1}{26}\begin{bmatrix} 7 & 3 \\ 3 & 5 \end{bmatrix}$ gives the same currents, and, being computed once, solves the problem for *any* source voltages: the columns of R^{-1} are the response to a unit source in each loop.

Check (energy): the source delivers $10 \cdot \frac{35}{13} = \frac{350}{13} \approx 26.9$ W and the resistors dissipate $2I_1^2 + 3(I_1 - I_2)^2 + 4I_2^2 = \frac{4550}{169} = \frac{350}{13}$ W.

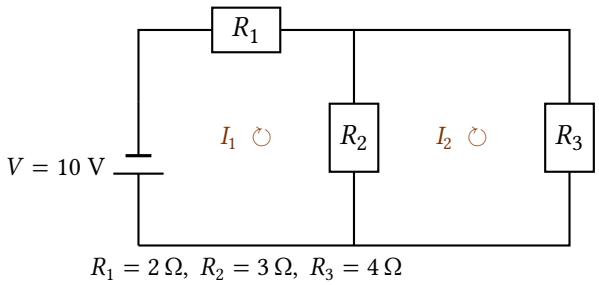


Figure 2.7: A two-loop resistor network with one source. The middle resistor R_2 carries the difference $I_1 - I_2$ of the two mesh currents.

2.4.4 Structures and Vibrations: the Stiffness Matrix

In structural analysis the displacements $\mathbf{x}$ of the nodes under loads $\mathbf{F}$ satisfy $K\mathbf{x} = \mathbf{F}$ with a symmetric **stiffness matrix** K. The structure is stable (every load gives a unique displacement) exactly when $\det K \neq 0$. A zero determinant means the structure is a *mechanism*: it can move without any spring stretching.

Example: Two Masses on Springs

For Figure 2.8 the equilibrium equations are $(k_1 + k_2)x_1 - k_2 x_2 = 0$ and $-k_2 x_1 + k_2 x_2 = F$, i.e.

$$K = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}, \qquad \det K = k_1 k_2 > 0.$$

With $k_1 = k_2 = 100$ N/m and $F = 10$ N: $K = \begin{bmatrix} 200 & -100 \\ -100 & 100 \end{bmatrix}$, $\det K = 10\,000$,

$K^{-1} = \frac{1}{10\,000}\begin{bmatrix} 100 & 100 \\ 100 & 200 \end{bmatrix}$, so

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = K^{-1}\begin{bmatrix} 0 \\ 10 \end{bmatrix} = \begin{bmatrix} 0.1 \\ 0.2 \end{bmatrix} \text{ m}.$$

If the wall is removed (a free body with two masses and one spring), $K = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$ has $\det K = 0$. Physically, the whole system can translate freely. The equations then have no solution for a net force $F \neq 0$ (the body accelerates) and infinitely many (a shifted copy) for balanced loads such as $F = (5, -5)$.

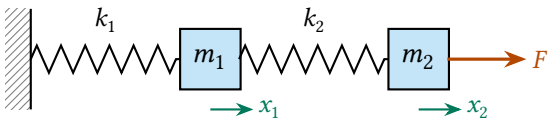


Figure 2.8: Two masses connected by springs to a fixed wall; x_1, x_2 are the displacements from equilibrium and F acts on the second mass.

Preview: Determinants Give Vibration Frequencies

For free vibrations of the two-mass system ($m_1 = m_2 = 1$ kg, $k_1 = k_2 = 100$ N/m) one looks for solutions that vary as $\sin \omega t$. They exist only when $K - \omega^2 I$ is *singular*:

$$\det(K - \lambda I) = (200 - \lambda)(100 - \lambda) - 10\,000 = \lambda^2 - 300\lambda + 10\,000 = 0, \qquad \lambda = \omega^2 = 150 \pm 50\sqrt{5},$$

so $\omega = 5(\sqrt{5} \pm 1) \approx 16.18$ and 6.18 rad/s. The condition $\det(A - \lambda I) = 0$ is the *characteristic equation*; it is the heart of the next topic, eigenvalues and eigenvectors.

2.4.5 Curve Fitting: the Vandermonde Determinant

To pass a polynomial of degree 2, $p(x) = ax^2 + bx + c$, through three points (x_i, y_i) we must solve a 3×3 system whose coefficient matrix is the **Vandermonde matrix**

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix}, \qquad \det V = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

The determinant is nonzero exactly when the three x-values are *distinct*, which is why a unique interpolating parabola exists in that case (and not if two sample points share the same x). For $(1, 2)$, $(2, 3)$, $(3, 6)$: $\det V = 1 \cdot 2 \cdot 1 = 2$ and the solution is $p(x) = x^2 - 2x + 3$, see Figure 2.9.

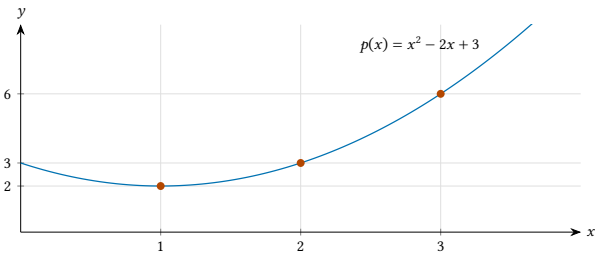


Figure 2.9: The unique parabola through $(1,2)$, $(2,3)$, $(3,6)$. The Vandermonde determinant $2 \neq 0$ guarantees that the three coefficients are determined uniquely by the three points.

2.5 Facts and Curiosities

1. Rows summing to zero force $\det = 0$**.** If every row of A sums to 0, then $A\mathbf{1} = \mathbf{0}$ for $\mathbf{1} = (1, \dots, 1)^T$, so A cannot be invertible. This is why the stiffness matrix of a free structure and the *Laplacian matrix* of a network are always singular.

2. The size of $\det A$ **does not measure "how invertible"** A **is.** The matrix $0.1\, I_{10}$ has $\det = 10^{-10}$ yet is perfectly well behaved ($A^{-1} = 10\, I$). Conversely $\begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}$ has $\det = 10^{-4}$ and is *nearly singular*: $x + y = 2$, $x + 1.0001y = 2$ gives $(x, y) = (2, 0)$, while changing the right side to 2.0001 gives $(1, 1)$: a change of 10^{-4} in one right-hand side moves the answer by 1. Engineers measure this sensitivity by the *condition number*, not by the determinant.

3. Area-preserving does not mean shape-preserving. The shear $\begin{bmatrix} 1 & 100 \\ 0 & 1 \end{bmatrix}$ has determinant 1 (areas unchanged) but stretches lengths enormously. Only *orthogonal* matrices preserve lengths and angles; they have $\det = \pm 1$.

4. Integer matrices with integer inverses. If A and A^{-1} both have integer entries then $\det A \cdot \det A^{-1} = 1$ forces $\det A = \pm 1$. Conversely, $\det A = \pm 1$ makes $A^{-1} = \pm \operatorname{adj}(A)$ integral. Such *unimodular* matrices (e.g. $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ with inverse $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$) map the integer lattice onto itself, and are used in crystallography and in lattice-based cryptography.

5. A short history. Determinants appeared in Japan in the 1680s (Seki Takakazu) and, independently, in Leibniz's letters of 1693. Cramer published his rule in 1750, and Sarrus found his diagonal scheme in 1833. They were used for solving equations long before matrices became a language of their own in the 19th century.

2.6 Worked Problems

Problem 1: Sarrus' Rule

Compute $\det(A)$ using the basket-weave method, where $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 0 & 5 \\ 1 & 2 & 6 \end{bmatrix}$.

Solution. Augmented matrix: $\begin{bmatrix} 2 & 1 & 3 & 2 & 1 \\ 4 & 0 & 5 & 4 & 0 \\ 1 & 2 & 6 & 1 & 2 \end{bmatrix}$. Downward: $2 \cdot 0 \cdot 6 = 0$, $1 \cdot 5 \cdot 1 = 5$, $3 \cdot 4 \cdot 2 = 24$; sum 29. Upward: $1 \cdot 0 \cdot 3 = 0$, $2 \cdot 5 \cdot 2 = 20$, $6 \cdot 4 \cdot 1 = 24$; sum 44. Hence $\det(A) = 29 - 44 = \mathbf{-15}$.

Problem 2: Row Replacement

Compute $\det A$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 1 & 2 & 4 \end{bmatrix}$.

Solution. Replacements (Property 3) do not change det. $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$ give

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

upper triangular, so $\det A = 1 \cdot 1 \cdot 1 = \mathbf{1}$.

Problem 3: 4×4 by Laplace Expansion

Compute $\det(A)$ for $A = \begin{bmatrix} 1 & 2 & 0 & 4 \\ 3 & 0 & 5 & 6 \\ 0 & 0 & 0 & 2 \\ 7 & 8 & 0 & 9 \end{bmatrix}$.

Solution. Column 3 has a single nonzero entry $a_{23} = 5$, so $\det(A) = 5\,(-1)^{2+3} M_{23}$ with

$$M_{23} = \begin{vmatrix} 1 & 2 & 4 \\ 0 & 0 & 2 \\ 7 & 8 & 9 \end{vmatrix}.$$

Expand M_{23} along row 2 (only the entry 2 at position $(2,3)$ survives):

$$M_{23} = 2\,(-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = -2(8 - 14) = 12, \qquad \det(A) = 5 \cdot (-1) \cdot 12 = \mathbf{-60}.$$

Problem 4: Scaling and Proportional Rows

Compute $\det A$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 8 & 10 & 12 \end{bmatrix}$.

Solution. Row 3 equals 2·row 2. Factoring 2 out of row 3 (Property 2) gives

$$\det A = 2 \det \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 4 & 5 & 6 \end{bmatrix},$$

and the new matrix has two equal rows, so its determinant is 0 (Property 6). Hence $\det A = \mathbf{0}$.

Problem 5: When Is the Determinant Zero?

Find all x for which $\det \begin{bmatrix} x & 1 & 2 \\ 3 & x & 4 \\ 5 & 6 & x \end{bmatrix} = 0$.

Solution. Expanding along the first row,

$$\det = x\begin{vmatrix} x & 4 \\ 6 & x \end{vmatrix} - \begin{vmatrix} 3 & 4 \\ 5 & x \end{vmatrix} + 2\begin{vmatrix} 3 & x \\ 5 & 6 \end{vmatrix} = x(x^2 - 24) - (3x - 20) + 2(18 - 5x) = x^3 - 37x + 56.$$

The cubic $x^3 - 37x + 56 = 0$ has no rational root (the candidates $\pm 1, \pm 2, \pm 4, \pm 7, \pm 8, \ldots$ all fail), but it changes sign three times, so it has **three real irrational roots**, approximately $x \approx -6.73,\ 1.63,\ 5.10$ (Figure 2.10). For exactly these values the matrix is singular.

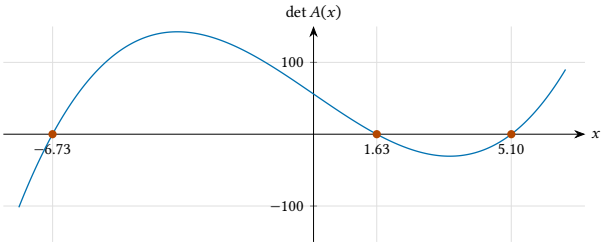


Figure 2.10: The determinant of Problem 5 as a function of x. Each of the three zero crossings is a value of x for which the matrix is singular.

Problem 6: A Parameter Makes the Matrix Singular

For which k is $A = \begin{bmatrix} 1 & k & 2 \\ 0 & 1 & k \\ k & 0 & 1 \end{bmatrix}$ not invertible?

Solution. Expanding along column 1:

$$\det A = 1\begin{vmatrix} 1 & k \\ 0 & 1 \end{vmatrix} + k\begin{vmatrix} k & 2 \\ 1 & k \end{vmatrix} = 1 + k(k^2 - 2) = k^3 - 2k + 1 = (k - 1)(k^2 + k - 1).$$

So A is singular for $k = 1$ and $k = \dfrac{-1 \pm \sqrt{5}}{2}$ ($\approx 0.618,\ -1.618$), and invertible for every other k.

Problem 7: Inverse by the Adjugate

Find A^{-1} for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 0 & 1 \end{bmatrix}$.

Solution. $\det A = 1(1-0) - 2(0-8) + 3(0-2) = 11 \neq 0$. The cofactors are

$$C = \begin{bmatrix} +\begin{vmatrix}1&4\\0&1\end{vmatrix} & -\begin{vmatrix}0&4\\2&1\end{vmatrix} & +\begin{vmatrix}0&1\\2&0\end{vmatrix} \\ -\begin{vmatrix}2&3\\0&1\end{vmatrix} & +\begin{vmatrix}1&3\\2&1\end{vmatrix} & -\begin{vmatrix}1&2\\2&0\end{vmatrix} \\ +\begin{vmatrix}2&3\\1&4\end{vmatrix} & -\begin{vmatrix}1&3\\0&4\end{vmatrix} & +\begin{vmatrix}1&2\\0&1\end{vmatrix} \end{bmatrix} = \begin{bmatrix} 1 & 8 & -2 \\ -2 & -5 & 4 \\ 5 & -4 & 1 \end{bmatrix}, \quad \operatorname{adj} A = C^T = \begin{bmatrix} 1 & -2 & 5 \\ 8 & -5 & -4 \\ -2 & 4 & 1 \end{bmatrix},$$

$$A^{-1} = \frac{1}{11}\begin{bmatrix} 1 & -2 & 5 \\ 8 & -5 & -4 \\ -2 & 4 & 1 \end{bmatrix}.$$

Problem 8: Linear Matrix Equation

Solve $AX + B = C$ for X, where $A = \begin{bmatrix}1&2\\3&4\end{bmatrix}$, $B = \begin{bmatrix}1&0\\0&1\end{bmatrix}$, $C = \begin{bmatrix}5&6\\7&8\end{bmatrix}$.

Solution. $AX = C - B = \begin{bmatrix}4&6\\7&7\end{bmatrix}$, so $X = A^{-1}(C-B)$. With $\det A = -2$,

$$A^{-1} = \frac{1}{-2}\begin{bmatrix}4&-2\\-3&1\end{bmatrix} = \begin{bmatrix}-2&1\\ \frac{3}{2}&-\frac{1}{2}\end{bmatrix}, \qquad X = \begin{bmatrix}-2&1\\ \frac{3}{2}&-\frac{1}{2}\end{bmatrix}\begin{bmatrix}4&6\\7&7\end{bmatrix} = \begin{bmatrix}-8+7 & -12+7\\ 6-\frac{7}{2} & 9-\frac{7}{2}\end{bmatrix} = \begin{bmatrix}-1&-5\\ \frac{5}{2}&\frac{11}{2}\end{bmatrix}.$$

Check: $AX = \begin{bmatrix}-1+5 & -5+11\\ -3+10 & -15+22\end{bmatrix} = \begin{bmatrix}4&6\\7&7\end{bmatrix}$. ☒

2.7 Conceptual & Computational Exercises

2.7.1 Conceptual Questions

Answer by *thinking*, with little or no calculation; give a reason or a counterexample.

1. **True or false?**
   (a) $\det(A+B) = \det(A) + \det(B)$.
   (b) If $\det A = 0$, then A has a row of zeros.
   (c) $\det(AB) = \det(BA)$ for $n \times n$ matrices.
   (d) If every row of A sums to 0, then $\det A = 0$.
   (e) If $A^2 = 0$, then A is not invertible.
   (f) If A and B are both invertible $n \times n$ matrices, then $A + B$ is invertible.
2. Why does the "diagonal scheme" of Sarrus not work for 4×4 matrices?
3. A 3×3 matrix has $\det A = 5$. Find the determinant after $R_1 \leftrightarrow R_2$, then $R_3 \to 4R_3$, then $R_2 \to R_2 + 7R_1$.
4. If A is $n \times n$ with $\det A = 3$, find $\det(-A)$ for $n = 3$ and for $n = 4$.
5. The columns of a 2×2 matrix are $\mathbf{a}_1 = (2, 1)$ and $\mathbf{a}_2 = (1, t)$. For which t is the matrix singular? Interpret the answer geometrically.
6. A matrix maps the unit square to a parallelogram of area 6 and reverses orientation. What is its determinant? What is the determinant of its inverse?
7. Show that an orthogonal matrix ($A^TA = I$) has $\det A = \pm 1$, and give one example of each sign.
8. If AB is singular, must A or B be singular? Why?

9. Is the inverse of a symmetric invertible matrix symmetric?
10. A triangular matrix has diagonal entries $2, -1, 0, 5$. Is it invertible? Why?
11. The system $A\mathbf{x} = \mathbf{b}$ with $\det A = 0$ was found to have the solution $\mathbf{x} = (1, 2)$. Can it be the *only* solution?
12. Why is $(AB)^{-1} = B^{-1}A^{-1}$ and not $A^{-1}B^{-1}$? Illustrate with "socks and shoes".

2.7.2 Computational Problems

1. Compute the determinants with Sarrus' rule: $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 5 & 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & -1 \\ 4 & 2 & 0 \end{bmatrix}$.
2. Compute the determinants using row operations and the properties of determinants:
   $C = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 5 & 6 \\ 0 & 3 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 4 \\ 0 & 5 & 2 \end{bmatrix}$.
3. Compute the determinants using Laplace expansion:
   $E = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 4 & 2 \\ 3 & 0 & 1 & 1 \\ 2 & 1 & 0 & 2 \end{bmatrix}$, $F = \begin{bmatrix} 2 & 1 & 0 & 1 \\ 1 & 3 & 2 & 0 \\ 0 & 2 & 1 & 4 \\ 1 & 0 & 3 & 2 \end{bmatrix}$.
4. Find the inverses with the adjugate formula, and check with $AA^{-1} = I$:
   $G = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 4 \\ 2 & 0 & 1 \end{bmatrix}$, $H = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix}$.
5. Solve $AX + B = C$ for X, where $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 7 & 5 \\ 9 & 4 \end{bmatrix}$.
6. Find all x with $\det \begin{bmatrix} x & 2 & 0 \\ 1 & x & 1 \\ 0 & 2 & x \end{bmatrix} = 0$.
7. For which values of k is $\begin{bmatrix} k & 1 & 0 \\ 1 & k & 1 \\ 0 & 1 & k \end{bmatrix}$ singular? (These are exactly the k for which this tridiagonal matrix is not invertible; the same pattern appears in vibrating chains of masses.)
8. Let $B = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & 2 & 3 \end{bmatrix}$. Compute B^{-1} and verify $BB^{-1} = I$. Then solve $BX = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix}$ for X.
9. Find the inverse of $T = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$ by Gauss–Jordan elimination.
10. The triangle with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) has area $\tfrac{1}{2}\left|\det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}\right|$. (a) Use this to find the area of the triangle with vertices $(2, 1)$, $(6, 2)$, $(3, 6)$. (b) Are $(0, 1)$, $(2, 5)$, $(3, 7)$ collinear? (Collinear points give a triangle of area 0.)
11. A 3×3 matrix A has $\det A = -4$. Find $\det(2A)$, $\det(A^{-1})$, $\det(A^2)$, $\det(A^{T}A^{-1})$ and $\det(\operatorname{adj} A)$.
12. A chain wall–spring–m_1–spring–m_2–spring–wall has three springs of stiffness 100 N/m. A force 10 N acts on m_1 only. Set up $K\mathbf{x} = \mathbf{F}$ and solve for the displacements.

13. The mesh equations of a circuit are $4I_1 - I_2 = 6$ and $-I_1 + 3I_2 = 0$. Solve for the currents by inverting the 2×2 matrix.

14. Find the quadratic $p(x) = ax^2 + bx + c$ through $(0, 1)$, $(1, 3)$, $(2, 7)$ and verify that the Vandermonde determinant is nonzero.

3

Systems of Linear Equations

Learning Objectives

After studying this chapter you should be able to:

- recognize linear equations and write a system in matrix and augmented-matrix form;
- interpret a system geometrically (lines in $\mathbb{R}^2$, planes in $\mathbb{R}^3$) and explain why it has no solution, exactly one solution, or infinitely many solutions;
- solve square systems by the inverse method and by Cramer's rule;
- solve any system by Gaussian and Gauss–Jordan elimination and describe the solution set with free parameters;
- classify a system using the rank of its matrices (Kronecker–Capelli theorem) and solve homogeneous systems;
- model problems from circuits, statics, chemistry, traffic and heat flow as linear systems, and interpret free variables and inconsistency physically.

When you first study mathematics, the equations you encounter are usually neat, orderly, and linear—meaning they follow predictable rules where causes and effects are directly proportional. However, the real world of physics and engineering is rarely so convenient; from the turbulent airflow over an aircraft wing to the rhythmic swing of a pendulum, nature is fundamentally nonlinear. While these complex equations are notoriously difficult and often impossible to solve using standard formulas, they govern how our universe truly operates. Even so, there are plenty of important linear equations as well, and to build the foundation needed for these advanced topics, we are going to learn the linear ones first. In this chapter, we study systems of linear equations.

Imagine a bakery that sells two types of cookies: chocolate chip cookies and oatmeal raisin cookies. The owner wants to figure out how many of each type to bake to meet a customer's order. A customer places an order for a total of 100 cookies.

- We'll let x be the number of chocolate chip cookies.
- We'll let y be the number of oatmeal raisin cookies.

The first piece of information gives us our first equation

$$x + y = 100.$$

The customer also specifies that the total cost of the order should be exactly \$130. Suppose chocolate chip cookies cost \$1.50 each and oatmeal raisin cookies cost \$1.00 each. This gives us our second equation:

$$1.50x + 1.00y = 130.$$

We now have two equations with two variables, a system of linear equations

$$\begin{cases} x + y = 100 \\ 1.50x + 1.00y = 130. \end{cases}$$

By solving this system, we can determine the exact number of chocolate chip and oatmeal raisin cookies needed to satisfy the customer's order. For instance, substituting $y = 100 - x$ into the second equation gives $1.50x + 100 - x = 130$, that is $0.50x = 30$, so $x = 60$ and $y = 40$: the bakery must bake 60 chocolate chip and 40 oatmeal raisin cookies.

Geometrically, each equation describes a straight line in the xy-plane, and the solution is the point where the two lines cross (Figure 3.1). This picture will be the key to understanding all the possible outcomes of a system of linear equations.

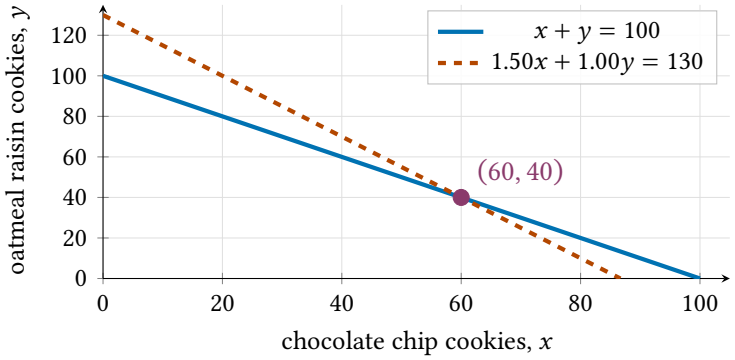


Figure 3.1: The order is the intersection point of two lines: 60 chocolate chip and 40 oatmeal raisin cookies.

Intuition. Why study systems of equations? Most real problems contain several unknown quantities that are tied together by several conditions: the unknown currents in a circuit, the unknown tensions in the cables of a structure, the unknown amounts of ingredients in a recipe. Each condition is one equation. *Linear* systems are the case that we can solve *completely and always*: by the end of the chapter you will be able to decide, for any such system, whether it has no solution, exactly one, or infinitely many, and to find them all. Even for nonlinear problems, engineers usually replace the true equations by linear ones near the point of interest (linearization), so this chapter is the workhorse behind much of applied mathematics.

3.1 Definition of a Linear Equation

Definition

A **linear equation** in n variables $x_1, x_2, \dots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where $a_1, a_2, \dots, a_n$ are constants (called coefficients), b is a constant, and the variables $x_1, x_2, \dots, x_n$ appear only to the first power with no products, powers, or other functions applied to them.

- Equation 1: $2x + 3y - 5z = 7$ is **linear** because it involves only first-degree terms in x, y, z with constant coefficients and no products or other functions.
- Equation 2: $x^2 + y - z = 4$ is **nonlinear** because of the x^2 term, which has degree 2.
- Equation 3: $4x - 2y + \sin(z) = 0$ is **nonlinear** because the $\sin(z)$ term involves a trigonometric function.
- Equation 4: $xy + 3z = 1$ is **nonlinear** because the xy term is a product of variables.
- Equation 5: $5x - 4y + 0z = 2$ is **linear** because it involves only first-degree terms, even though the coefficient of z is 0.

Common mistake. Linear in the unknowns, not in the numbers. What matters is how the *variables* enter. The equation $\pi x + \sqrt{2}\, y = e$ is linear (the coefficients $\pi, \sqrt{2}$ and the constant e are just numbers), but $xy = 1$, $x^2 + y = 0$, $\sin x + y = 0$ and $\frac{1}{x} + y = 3$ are not. A product of two unknowns is not allowed even when it is hidden: $z(x+1) = 2$ contains the term zx.

A **system of linear equations** with m equations in n variables $x_1, x_2, \dots, x_n$ is a collection of m equations, each of the form

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i, \quad i = 1, 2, \dots, m,$$

where a_{ij} are the **coefficients**, b_i are the **constants**, and $x_1, x_2, \dots, x_n$ are the **variables**. Symbolically, the system can be written as follows

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

This system can be represented in **matrix form** as follows $AX = B$, where,

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

are the $m \times n$ **coefficient matrix**, **variable matrix**, and **constant matrix**, respectively. The **augmented matrix** of the system is the $m \times (n+1)$ matrix

$$[A \mid B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right].$$

Why we need it. *When several conditions must hold at once we need a name for them as a package (a system) and a compact notation. The matrix form $AX = B$ looks like the equation $ax = b$, so everything we know about matrices can be reused. The augmented matrix stores all the numbers of the system and hides the variable names, which is exactly what elimination needs.*

A **solution** to a system of linear equations with m equations in n variables $x_1, x_2, \dots, x_n$ is a set of values $(s_1, s_2, \dots, s_n)$ that satisfies each equation in the system when substituted for $x_1, x_2, \dots, x_n$. The **solution set** is the collection of all such solutions that satisfy the system simultaneously. A system that has at least one solution is called **consistent**; a system that has no solution is called **inconsistent**. The possible outcomes for the solution set of a system of linear equations are as follows

- **Unique Solution**: The system has exactly one solution, corresponding to a single point in $\mathbb{R}^n$.
- **Infinitely Many Solutions**: The system has an infinite number of solutions, often described by one or more free variables.
- **No Solution**: The system is inconsistent.

Why can a system never have exactly two solutions? Suppose $X_1 \neq X_2$ are two different solutions of $AX = B$, so $AX_1 = B$ and $AX_2 = B$. For every real number t, $X_1 + t(X_2 - X_1)$ is also a solution, because

$$A\big(X_1 + t(X_2 - X_1)\big) = (1-t)\,AX_1 + t\,AX_2 = (1-t)B + tB = B.$$

These solutions are all different (they fill the whole line through X_1 and X_2), so a system with two solutions automatically has infinitely many. Hence there are exactly three possibilities: no solution, one solution, or infinitely many solutions.

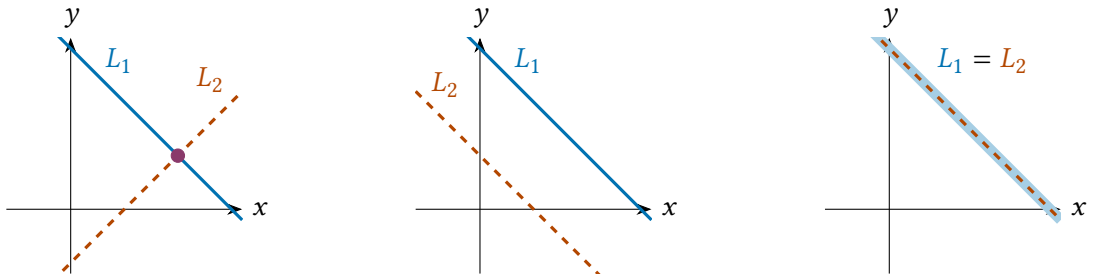
3.1.1 Geometric Interpretation

A linear equation $ax + by = c$ (with a, b not both zero) is a *line* in the plane, and a linear equation $ax + by + cz = d$ (with a, b, c not all zero) is a *plane* in space. A solution of a system is a point that lies on all these lines (planes) at the same time, so

$$\text{solving a system} \;=\; \text{finding the common intersection of lines or planes.}$$

Why we need it. *Pictures tell us in advance what can happen (a point, a line, nothing) and make the algebraic results, such as the rank theorem, easy to remember.*

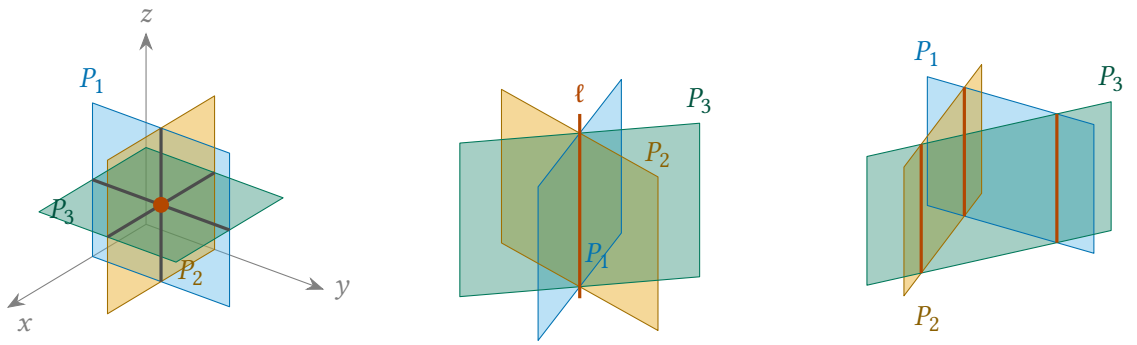
Two equations in two unknowns. Two lines in the plane can be positioned in exactly three ways (Figure 3.2): they cross in one point, they are parallel and distinct, or they coincide.



(a) One common point **(b)** No common point **(c)** All points common

Figure 3.2: Two equations in two unknowns. (a) $x + y = 3,\ x - y = 1$: the lines cross at $(2, 1)$, the *unique solution*. (b) $x + y = 3,\ x + y = 1$: parallel lines, *no solution*. (c) $x + y = 3,\ 2x + 2y = 6$: the same line, *infinitely many solutions* $(t,\ 3 - t)$, $t \in \mathbb{R}$.

Three equations in three unknowns. Now every equation is a plane. Figure 3.3 shows the three typical situations, one for each type of solution set.



(a) One common point **(b)** A common line **(c)** Pairwise crossings only

Figure 3.3: Three equations in three unknowns (each equation is a plane P_i). (a) The planes meet in a single point, the *unique solution* (for example $x = 1,\ y = 1,\ z = 1$). (b) The planes share a common line ℓ, so there are *infinitely many solutions* with one free parameter (for example $x = 0,\ y = 0,\ x + y = 0$: the line ℓ is the z-axis). (c) Every two planes meet in a line, but the three lines are parallel and the planes have no point in common (a "triangular prism"): *no solution* (for example $x = 0,\ y = 0,\ x + y = 1$).

These are not the only possible pictures. For instance, two planes may be parallel (no solution), or all three planes may coincide (a whole plane of solutions). The precise classification in terms of ranks (see Section 3.2.4) is summarized in Table 3.1.

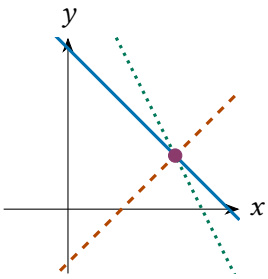
> **Common mistake. Pairwise intersections are not enough.** In Figure 3.3(c) every two planes meet in a line, yet the three planes have no common point (the "triangular prism"). Checking pairs is never a substitute for solving the whole system. Likewise, *two* equations in three unknowns can never have a *unique* solution: two planes meet in a line or not at all.

rank(A)	rank($[A\|B]$)	Position of the three planes	Solutions
3	3	the planes meet in exactly one point	unique
2	2	the planes share a common line (or two planes coincide and the third crosses them)	infinitely many, 1 parameter
1	1	all three planes coincide	infinitely many, 2 parameters
2	3	a triangular prism, or two parallel planes cut by a third one	none
1	2	all planes parallel, at least two of them distinct	none

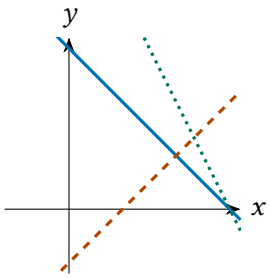
Table 3.1: Relative position of three planes $a_i x + b_i y + c_i z = d_i$, $i = 1, 2, 3$, and the type of solution set.

3.1.2 More Equations Than Unknowns

A system with more equations than unknowns ($m > n$) is called **overdetermined**: it asks more of the unknowns than they can usually give. In the plane (Figure 3.4), two lines already fix a point, and a third line must happen to pass through exactly that point.



(a) Consistent: three lines through $(2, 1)$



(b) Inconsistent: a small triangle, no common point

Figure 3.4: Three equations in two unknowns. (a) $x + y = 3$, $x - y = 1$, $2x + y = 5$ all pass through $(2, 1)$: the extra equation is redundant. (b) Replacing the last equation by $2x + y = 6$ makes the lines enclose a triangle with vertices $(2, 1)$, $(3, 0)$, $(\frac{7}{3}, \frac{4}{3})$: no point lies on all three.

Common mistake. Counting equations does not decide the solution type. Three equations in three unknowns can still have no unique solution if one equation is a combination of the others, and four equations in three unknowns can have a unique solution if one of them is redundant (see the four-equation example in Section 3.2.3). What decides is the *rank*, not the number of equations.

3.2 Methods for Solving Systems of Linear Equations

There are several ways to solve systems of linear equations. They can be introduced by grouping them into broad families:

- **Elementary algebraic methods**: substitution, elimination, and Cramer's rule.
- **Direct matrix methods**: Gaussian elimination, LU decomposition, Cholesky decomposition, QR decomposition, the matrix inverse method, and SVD/pseudoinverse techniques.
- Least-squares methods for overdetermined or inconsistent systems: the normal equations, QR least squares, and SVD-based least squares.
- **Iterative methods** for large or sparse systems: the Jacobi method, the Gauss–Seidel method, successive over-relaxation (SOR), the conjugate gradient (CG) method, Krylov subspace methods

such as BiCGSTAB, GMRES, and MINRES, multigrid methods, and preconditioned iterative methods.

- **Specialized and structured methods**: the Thomas algorithm for tridiagonal systems, fast Fourier/cyclic reduction, sparse direct solvers, and block and partitioned methods.

Thus, the choice of method depends on the size, structure, conditioning, and type of the linear system. We will study the following methods: the Inverse Method, Cramer's Rule, and Gaussian Elimination and Gauss–Jordan Methods. Each method provides a systematic approach to finding the solution set, with varying advantages depending on the system's properties. Table 3.2 gives an overview.

Method	Applies to	Requires	Best used when
Inverse method	square systems ($m = n$)	$\det(A) \neq 0$	several right-hand sides share the same A; theoretical arguments
Cramer's rule	square systems	$\det(A) \neq 0$	very small systems (2×2, 3×3); formulas that contain parameters
Gaussian elimination	any $m \times n$ system	nothing	general purpose; detects all three types of solution sets
Gauss–Jordan elimination	any $m \times n$ system	nothing	reading the solution directly from the RREF

Table 3.2: Comparison of the methods for solving $AX = B$.

3.2.1 Inverse Method

The Inverse Method applies to a system of n linear equations in n variables, written in matrix form as $AX = B$, where A is the $n \times n$ coefficient matrix, X is the variable matrix, and B is the constant matrix. If A is invertible (i.e., $\det(A) \neq 0$), multiplying both sides of $AX = B$ from the left by A^{-1} gives $A^{-1}AX = A^{-1}B$, that is,

$$X = A^{-1}B.$$

Since A^{-1} is unique, the solution is unique.

Intuition. Undo what A did. $AX = B$ says "A turns the unknown X into the known B". The inverse matrix is the machine that *undoes* A, so applying it to B recovers X. This is the matrix version of dividing $ax = b$ by a: there is no "matrix division", but multiplying by A^{-1} plays the same role.

Common mistake. Multiply on the correct side, and do not invert needlessly. From $AX = B$ one gets $X = A^{-1}B$, never $X = BA^{-1}$ (that would solve $XA = B$, and the sizes usually do not even match). Also, computing A^{-1} just to solve *one* system is wasteful: elimination solves it directly with less work and fewer rounding errors on a computer. The inverse pays off by hand when the same A must be used with several right-hand sides.

Pros:

- Provides a direct formula for the solution when A is invertible.
- Useful for systems where A^{-1} is already known or needed for multiple systems with the same A.

Cons:

- Requires computing A^{-1}, which can be computationally lengthy for large n.
- Only applicable when A is square and invertible, limiting its use for systems with no unique solution or non-square matrices.

Example: Solving a System Using the Inverse Method

Solve the system of linear equations using the inverse method.

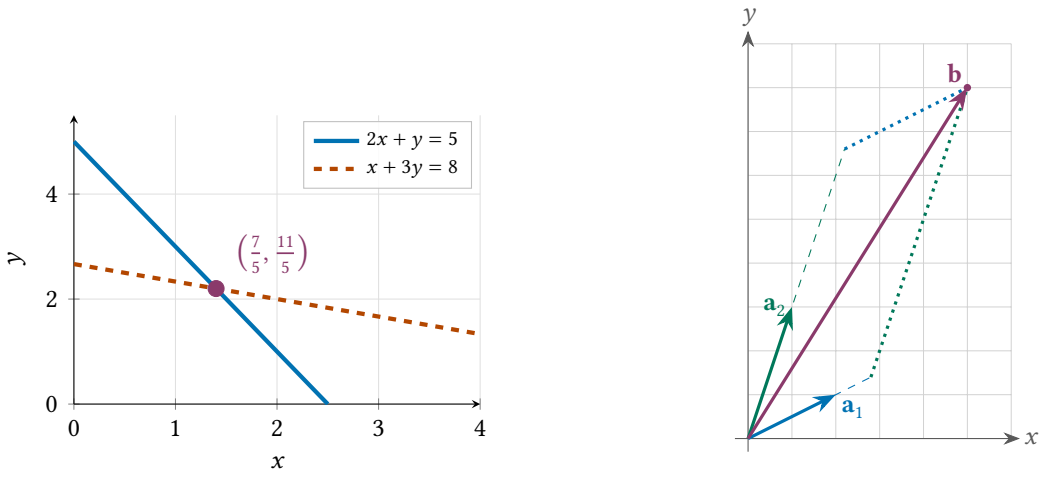
$$\begin{cases} 2x + y = 5 \\ x + 3y = 8 \end{cases}$$

Solution. Write the system in matrix form $AX = B$, where,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 8 \end{bmatrix}.$$

1. Compute $\det(A) = (2 \cdot 3) - (1 \cdot 1) = 6 - 1 = 5$. Since $\det(A) \neq 0$, A is invertible.
2. Find $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$.
3. Compute $X = A^{-1}B = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \cdot 5 + (-\frac{1}{5}) \cdot 8 \\ (-\frac{1}{5}) \cdot 5 + \frac{2}{5} \cdot 8 \end{bmatrix} = \begin{bmatrix} \frac{7}{5} \\ \frac{11}{5} \end{bmatrix}$.

Thus, the solution is $x = \frac{7}{5}$, $y = \frac{11}{5}$. *Check:* $2 \cdot \frac{7}{5} + \frac{11}{5} = \frac{25}{5} = 5$ and $\frac{7}{5} + 3 \cdot \frac{11}{5} = \frac{40}{5} = 8$. Geometrically, the solution is the point where the two lines meet (Figure 3.5).



(a) row picture: two lines meet

(b) column picture: $\frac{7}{5}\mathbf{a}_1 + \frac{11}{5}\mathbf{a}_2 = \mathbf{b}$

Figure 3.5: Two pictures of the same system $2x + y = 5$, $x + 3y = 8$. (a) Each equation is a line and the solution is their common point. (b) The columns $\mathbf{a}_1 = (2, 1)$, $\mathbf{a}_2 = (1, 3)$ and $\mathbf{b} = (5, 8)$: the system asks for the numbers x, y with $x\,\mathbf{a}_1 + y\,\mathbf{a}_2 = \mathbf{b}$. Stretching $\mathbf{a}_1$ by 1.4 and $\mathbf{a}_2$ by 2.2 makes the two pieces add up exactly to $\mathbf{b}$.

Intuition. Two ways to read $AX = B$. In the *row picture* every equation is a line (or plane) and the solution is where they all meet. In the *column picture* the same system says

$$x_1\,\mathbf{a}_1 + x_2\,\mathbf{a}_2 + \cdots + x_n\,\mathbf{a}_n = B,$$

where $\mathbf{a}_1, \ldots, \mathbf{a}_n$ are the columns of A: we are looking for the weights with which the columns must be combined to build B. A solution exists exactly when B can be built from the columns; it is unique when the columns are independent (only one way to build B); there are infinitely many solutions when the columns are dependent (several ways); and there is none when B lies outside everything the columns can reach. This second picture is the idea behind the rank criterion later in the chapter.

3.2.2 Cramer's Rule

Cramer's Rule provides a method to solve a system of n linear equations in n variables, written as $AX = B$, where A is an $n \times n$ coefficient matrix, X is the variable matrix, and B is the constant matrix. If $\det(A) \neq 0$, the system has a **unique solution**, and the solution for each variable x_i is given by

$$x_i = \frac{\det(A_i)}{\det(A)}, \quad i = 1, 2, \ldots, n,$$

where A_i is the matrix obtained by replacing the i-th column of A with B. For a system of two equations

$$\begin{cases} a_{11}x + a_{12}y = b_1 \\ a_{21}x + a_{22}y = b_2 \end{cases}$$

the rule gives

$$x = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}} = \frac{b_1a_{22} - a_{12}b_2}{a_{11}a_{22} - a_{12}a_{21}}, \qquad y = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}} = \frac{a_{11}b_2 - b_1a_{21}}{a_{11}a_{22} - a_{12}a_{21}}.$$

Cases for Solutions:

- If $\det(A) \neq 0$, the system has exactly one solution, computed using the formula above.
- If $\det(A) = 0$ and $\det(A_i) = 0$ for all i, the system may have **infinitely many solutions** or **no solution**, depending on consistency (determined by further analysis, e.g., row reduction).
- If $\det(A) = 0$ and $\det(A_i) \neq 0$ for at least one i, the system has **no solution** (inconsistent).

Common mistake. **Common mistakes with Cramer's rule.** Replace the *column* i of A (not a row) by B and leave the other columns where they are. Compute $\det A$ *first*: if it is 0, stop, because "$x_i = \det A_i/0$" means nothing; use elimination instead to decide between no solution and infinitely many. Finally, $\det A_i = 0$ for all i does not mean there is a solution: with $\det A = 0$ it only says that, if there is one, there are infinitely many.

Pros:

- Provides an explicit formula for each variable when a unique solution exists.
- Useful for small systems ($n = 2$ or 3) and theoretical understanding.

Cons:

- Requires computing $n + 1$ determinants, which is computationally intensive for large n (with cofactor expansion the work grows factorially with n).
- Inapplicable when $\det(A) = 0$, limiting its use for systems with no unique solution.

Example: Solving a System Using Cramer's Rule

Solve the system of linear equations

$$\begin{cases} 2x + y + z = 5 \\ x + 3y + 2z = 8 \\ 4x + y + 3z = 9 \end{cases}$$

using Cramer's Rule.

Solution. Write the system as $AX = B$, where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 3 & 2 \\ 4 & 1 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix}.$$

Compute $\det(A)$ using row 1: $\det(A) = 2(3 \cdot 3 - 2 \cdot 1) - 1(1 \cdot 3 - 2 \cdot 4) + 1(1 \cdot 1 - 3 \cdot 4) = 8 \neq 0$, so the system has a **unique solution**. Form A_1, A_2, A_3:

$$A_1 = \begin{bmatrix} 5 & 1 & 1 \\ 8 & 3 & 2 \\ 9 & 1 & 3 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 2 & 5 & 1 \\ 1 & 8 & 2 \\ 4 & 9 & 3 \end{bmatrix}, \qquad A_3 = \begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & 8 \\ 4 & 1 & 9 \end{bmatrix}.$$

Their determinants: $\det(A_1) = 5(3 \cdot 3 - 2 \cdot 1) - 1(8 \cdot 3 - 2 \cdot 9) + 1(8 \cdot 1 - 3 \cdot 9) = 10$,

$$\det(A_2) = 2(8 \cdot 3 - 2 \cdot 9) - 5(1 \cdot 3 - 2 \cdot 4) + 1(1 \cdot 9 - 8 \cdot 4) = 14,$$
$$\det(A_3) = 2(3 \cdot 9 - 8 \cdot 1) - 1(1 \cdot 9 - 8 \cdot 4) + 5(1 \cdot 1 - 3 \cdot 4) = 6.$$

Hence,

$$x = \frac{\det(A_1)}{\det(A)} = \frac{10}{8} = \frac{5}{4}, \qquad y = \frac{\det(A_2)}{\det(A)} = \frac{14}{8} = \frac{7}{4}, \qquad z = \frac{\det(A_3)}{\det(A)} = \frac{6}{8} = \frac{3}{4}.$$

Example: Nutrition Planning

A nutritionist wants to create a meal for a patient using three different food sources:

- **Food A:** 200 calories, 10g protein, 20g carbs
- **Food B:** 300 calories, 15g protein, 25g carbs
- **Food C:** 150 calories, 5g protein, 10g carbs

The nutritionist needs the total meal to contain exactly **1100 calories**, **50g of protein**, and **90g of carbs**. How many portions of each food source are needed?

Solution. Let x, y, and z be the number of portions of Foods A, B, and C, respectively. The nutritional requirements yield the system:

$$\begin{aligned} 200x + 300y + 150z = 1100 &\implies 4x + 6y + 3z = 22 \\ 10x + 15y + 5z = 50 &\implies 2x + 3y + z = 10 \\ 20x + 25y + 10z = 90 &\implies 4x + 5y + 2z = 18 \end{aligned}$$

We can now solve this system using Cramer's Rule. The coefficient matrix, A, contains the coefficients of the variables, and the constant vector, B, contains the numbers on the right side of the equals sign.

$$A = \begin{bmatrix} 4 & 6 & 3 \\ 2 & 3 & 1 \\ 4 & 5 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 22 \\ 10 \\ 18 \end{bmatrix}.$$

The determinant of the coefficient matrix equals

$$\det(A) = 4\begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} - 6\begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} + 3\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = 4(3 \cdot 2 - 1 \cdot 5) - 6(2 \cdot 2 - 1 \cdot 4) + 3(2 \cdot 5 - 3 \cdot 4) = -2.$$

Since $\det(A) = -2 \neq 0$, the system has a unique solution.

$$D_x = \begin{vmatrix} 22 & 6 & 3 \\ 10 & 3 & 1 \\ 18 & 5 & 2 \end{vmatrix} = 22(1) - 6(2) + 3(-4) = -2 \implies x = \frac{-2}{-2} = 1,$$

$$D_y = \begin{vmatrix} 4 & 22 & 3 \\ 2 & 10 & 1 \\ 4 & 18 & 2 \end{vmatrix} = 4(2) - 22(0) + 3(-4) = -4 \implies y = \frac{-4}{-2} = 2,$$

$$D_z = \begin{vmatrix} 4 & 6 & 22 \\ 2 & 3 & 10 \\ 4 & 5 & 18 \end{vmatrix} = 4(4) - 6(-4) + 22(-2) = -4 \implies z = \frac{-4}{-2} = 2.$$

The nutritionist needs **1 portion of Food A**, **2 portions of Food B**, and **2 portions of Food C** to meet the patient's nutritional requirements. *Check:* calories $200 + 600 + 300 = 1100$; protein $10 + 30 + 10 = 50$; carbs $20 + 50 + 20 = 90$.

3.2.3 Gaussian Elimination and Gauss–Jordan Methods

Two systems of linear equations are **equivalent** if they have the same solution set. The idea behind Gaussian Elimination and Gauss–Jordan Elimination is to transform a system of linear equations into an equivalent system that is easier to solve using **elementary row operations**.

Definition

Elementary Row Operations: The following operations on the augmented matrix of a system produce an **equivalent system**:

1. **Row Interchange**: Swap two rows, denoted as $R_i \leftrightarrow R_j$.
2. **Row Scaling**: Multiply a row by a *nonzero* scalar k, denoted as $kR_i \to R_i$.
3. **Row Addition**: Add a multiple of one row to a different row ($i \neq j$), denoted as $R_i + kR_j \to R_i$.

Two matrices A and B are **row-equivalent**, $A \sim B$, if one can be obtained from the other by a finite sequence of these operations. To track operations, we use the above notations beside each step. Performing these operations on an **augmented matrix** $[A|B]$ transforms it into a new augmented matrix corresponding to an equivalent system.

Intuition. **Why the solution set does not change.** Every elementary row operation can be undone by another elementary row operation: a swap is undone by the same swap, scaling by $k \neq 0$ is undone by scaling with $1/k$, and $R_i + kR_j \to R_i$ is undone by $R_i - kR_j \to R_i$. Therefore a solution of the old system is a solution of the new one and vice versa. This is also why scaling by $k = 0$ is forbidden: it cannot be undone, and it would destroy an equation. A combined step such as $3R_2 - 2R_1 \to R_2$ is allowed, because it is row scaling by $3 \neq 0$ followed by row addition (Figure 3.6).

Intuition. **Why elimination is allowed to "rotate the lines".** Every line through the intersection point of two lines is a combination of their equations. A row operation replaces an equation by such a combination, so the new line still passes through the same point. Elimination uses this freedom to rotate the lines until they are horizontal and vertical, where the intersection can simply be read off.

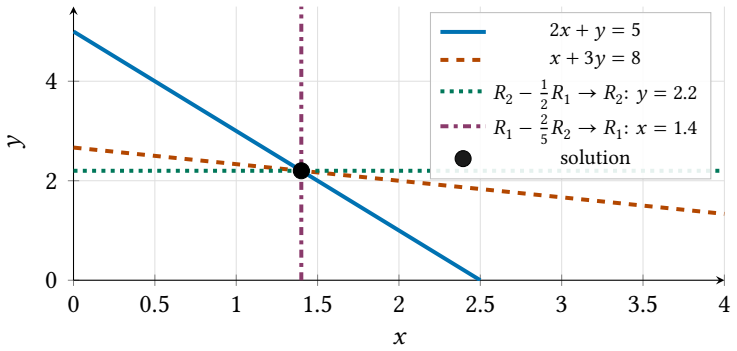


Figure 3.6: What elimination does to the picture. Starting from the lines $2x + y = 5$ and $x + 3y = 8$, the step $R_2 - \frac{1}{2}R_1 \to R_2$ replaces the second line by the horizontal line $2.5y = 5.5$, i.e. $y = 2.2$; then $R_1 - \frac{2}{5}R_2 \to R_1$ replaces the first line by the vertical line $2x = 2.8$, i.e. $x = 1.4$. All lines pass through the same point $(1.4, 2.2)$: row operations rotate the lines about the solution, which never moves.

Example: Elementary Row Operations

Consider the augmented matrix $[A|B] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 2 & 8 & 9 & 10 \end{array}\right]$. Apply one of each elementary row operation:

1. **Row Interchange** ($R_1 \leftrightarrow R_3$):
$$\left[\begin{array}{ccc|c} 2 & 8 & 9 & 10 \\ 0 & 5 & 6 & 7 \\ 1 & 2 & 3 & 4 \end{array}\right].$$
2. **Row Scaling** ($\frac{1}{5}R_2 \to R_2$) on the original matrix:
$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & \frac{6}{5} & \frac{7}{5} \\ 2 & 8 & 9 & 10 \end{array}\right].$$
3. **Row Addition** ($R_3 - 2R_1 \to R_3$) on the original matrix:
$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 4 & 3 & 2 \end{array}\right].$$

Definition: Row Echelon Form (REF) and Reduced Row Echelon Form (RREF)

A matrix is in **row echelon form** (REF) if

1. all rows consisting entirely of zeros are at the bottom;
2. the first nonzero entry of each nonzero row (its **leading entry** or **pivot**) is 1 (a **leading 1**);
3. the pivot of each nonzero row is to the right of the pivot in the row above it.

A matrix is in **reduced row echelon form** (RREF) if it is in REF and, in addition, each column containing a pivot has zeros in all other entries.

Why we need it. *Echelon form is the target of elimination: its staircase shape makes the system easy to solve from the bottom up, and in RREF the solution can be read off directly.*

These forms are achieved using **elementary row operations** and are used to solve **systems of linear equations** by transforming the **augmented matrix** into a simpler equivalent form. The pivots of a matrix in REF form a "staircase" going down and to the right (Figure 3.7).

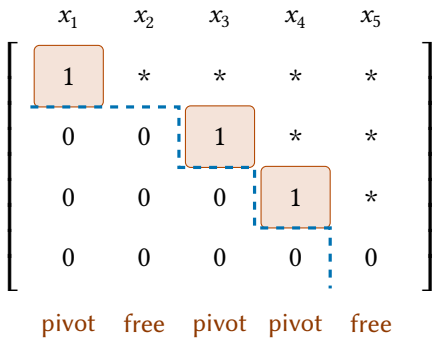


Figure 3.7: A matrix in row echelon form ($*$ = any number). The boxed entries are the pivots; below the dashed staircase everything is 0. Columns with a pivot correspond to pivot (basic) variables, columns without a pivot to free variables.

Intuition. **Why a staircase?** Each nonzero row starts further to the right than the row above, so the *last* nonzero row involves the fewest variables. Solve it first and substitute upward (*back-substitution*): the staircase is the shape of a system that is "already half solved".

Common mistake. **"Pivot" before and after scaling.** While eliminating, the entry you are about to use is called the pivot even before it has been scaled to 1 (the boxed entries in the worked examples, such as -1 or 5); the definition of REF asks for leading 1's, so scaling is the last step in each column. (Some books do not require leading 1's; the pivot columns, the rank and the solution set are the same either way.) Also remember that a pivot must be a *nonzero* entry, and that in a skipped column there is no pivot at all.

Example: Identifying Row Echelon and Reduced Row Echelon Forms

Consider the following matrices (pivots are boxed):

1. $\begin{bmatrix} \boxed{1} & 2 & 0 & 3 \\ 0 & \boxed{1} & 4 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in **row echelon form** (all pivots equal 1, pivots shift to the right, the zero row is at the bottom), but it is *not* in reduced row echelon form, because the entry 2 above the second pivot is not zero.
2. $\begin{bmatrix} \boxed{1} & 0 & 0 & 2 \\ 0 & \boxed{1} & 0 & 3 \\ 0 & 0 & \boxed{1} & 4 \end{bmatrix}$ is in **reduced row echelon form** (it is in REF, and every pivot column has zeros above and below the pivot).
3. $\begin{bmatrix} 0 & \boxed{1} & 2 \\ \boxed{1} & 0 & 3 \end{bmatrix}$ is *not* in row echelon form (the pivot in row 2 is to the left of the pivot in row 1).

Intuition. **RREF is unique, REF is not.** The **reduced row echelon form** (RREF) of a matrix is unique, meaning that every matrix has exactly one RREF, regardless of the sequence of elementary row operations used. However, the **row echelon form** (REF) is not unique; different sequences of operations may lead to different REF representations of the same matrix. Uniqueness is why the RREF is the "canonical" form: two matrices of the same size are row-equivalent exactly when they have the same RREF.

Definition

A **pivot column** is a column that contains a pivot. A **pivot variable**, also known as a **basic variable**, is a variable that corresponds to a pivot column in a matrix that is in row-echelon form (REF) or reduced row-echelon form (RREF).

A **free variable** in a **system of linear equations** is a variable that is not a **pivot variable** in the row-echelon form (REF) or reduced row-echelon form (RREF) of the augmented matrix. Free variables arise when there are fewer pivot columns than the number of variables; if the system is consistent, the free variables can take any value, leading to **infinitely many solutions**.

To solve a system of linear equations using Gaussian elimination, we apply the following algorithm.

Algorithm 1: Gaussian Elimination Method for the augmented matrix $[A|B]$ of an $m \times n$ system

Input: the augmented matrix $[A|B]$
Output: a row-echelon form of $[A|B]$ and the description of the solution set

```
r ← 1;                    // r is the row that receives the next pivot
for c ← 1 to n do
    if all entries of column c in rows r, r + 1, ..., m are 0 then
        skip column c (no pivot, so x_c is a free variable);
    else
        swap rows (if necessary) so that a nonzero entry is in position (r, c);
        scale row r so that this pivot equals 1;
        use row addition to create zeros below the pivot;
        r ← r + 1;
if some row has the form [ 0 ··· 0 | d ] with d ≠ 0 then
    the system is inconsistent: no solution;
else
    perform back-substitution, starting from the last nonzero row, and give a parameter to
      every free variable;
```

Common mistake. **Common mistakes in elimination.**

- Write down every step: errors hide in mental multi-step arithmetic.
- $R_i + kR_j \to R_i$ changes only row i; row j stays as it was. If you perform several operations in one step, use the *old* version of the row you are adding.
- Never multiply a row by 0 (this destroys an equation), and do not replace a row by a multiple of itself with factor 0, e.g. $R_2 - R_2 \to R_2$.
- Always check your answer in the *original* equations, not in the reduced ones: an early slip is carried into every later matrix, and the reduced system can look perfectly consistent.

If we continue the elimination process further, making zeros both *below and above* each pivot (instead of only below), the matrix will be transformed into the **reduced row-echelon form**. This extended procedure is called the **Gauss–Jordan Elimination Method**, and it allows the solution to be read directly from the final matrix without the need for back-substitution.

Numerical Practice

On a computer one usually chooses as pivot the entry of largest absolute value in the column (*partial pivoting*) instead of the first nonzero one, because this reduces rounding errors. For hand calculations with small integers, any convenient nonzero pivot is fine.

Below, we discuss the pros and cons of Gaussian Elimination and Gauss–Jordan Elimination, along with how to determine the type of solution: no solution, unique solution, or infinitely many solutions.

1. **Gaussian Elimination**:
   - **Pros**: Efficient for large systems, requires fewer operations than Gauss–Jordan (roughly two thirds as many for large n), handles all solution types (**unique**, **infinite**, or **no solution**).
   - **Cons**: The main disadvantage is the need for **back-substitution** to find the values of the variables. This process can be more prone to human error in manual calculations.
   - **Determining Solution Type**: Transform the augmented matrix to **row-echelon form**:
     - If a row is $[0 \cdots 0 \mid c]$ with $c \neq 0$, the system has **no solution**.
     - If consistent (no such row) and there are free variables (columns without a pivot among the first n columns), the system has **infinitely many solutions**.
     - If consistent and there are no free variables (every variable column has a pivot), the system has a **unique solution**.
   - **Finding the Solution Set for Infinitely Many Solutions**: In REF, express pivot variables in terms of free variables using back-substitution. Assign parameters (e.g., t, s) to free variables and write the solution as a parametric expression.
2. **Gauss–Jordan Elimination**:
   - **Pros**: Transforms to reduced row-echelon form (RREF), allowing direct reading of solutions, clear for identifying free variables.
   - **Cons**: Requires more row operations than Gaussian Elimination, less efficient for very large systems.
   - **Determining Solution Type**: Transform the augmented matrix to RREF:
     - If a row is $[0 \cdots 0 \mid c]$ with $c \neq 0$, the system has **no solution**.
     - If consistent and there are columns without a pivot, the system has **infinitely many solutions**.
     - If consistent and every variable column has a pivot (the coefficient part is the identity matrix I_n, followed by zero rows if $m > n$), the system has a **unique solution**.
   - **Finding the Solution Set for Infinitely Many Solutions**: In RREF, the solution is read directly. Free variables correspond to non-pivot columns. Assign parameters to free variables and express pivot variables as linear combinations of these parameters.

Common mistake. **A zero in the pivot position.** During **Gaussian Elimination** or **Gauss–Jordan Elimination**, if the target pivot element in a column is zero and all elements below it in that column are also zero, then that column cannot produce a pivot. In this case, skip to the next column (in the same row) to continue the elimination process. The variable belonging to the skipped column is a free variable.

Example: A Zero Pivot, Skipping a Column

Solve the system

$$\begin{cases} x + 2y + z = 4 \\ 2x + 4y + 3z = 9 \\ x + 2y + 2z = 5 \end{cases}$$

Solution. The augmented matrix is

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 4 \\ 2 & 4 & 3 & 9 \\ 1 & 2 & 2 & 5 \end{array}\right].$$

Step 1: Pivot in column 1 (the boxed 1). Eliminate below:

$$R_2 - 2R_1 \to R_2, \quad R_3 - R_1 \to R_3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Step 2: Look at column 2 in rows 2 and 3: both entries are 0, so this column *cannot* produce a pivot. We skip it (so y will be a free variable) and look for the pivot in column 3, row 2, which is already 1. Eliminate below:

$$R_3 - R_2 \to R_3$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The matrix is in row-echelon form. The pivots are in columns 1 and 3, so x and z are pivot variables and y is free (a free variable need not be the last one). The consistent system has infinitely many solutions.

Back-substitution: let $y = t, t \in \mathbb{R}$. From row 2: $z = 1$. From row 1: $x + 2t + 1 = 4$, so $x = 3 - 2t$. The general solution is

$$(x, y, z) = (3 - 2t,\ t,\ 1), \quad t \in \mathbb{R}.$$

Geometrically, the three planes share a common line, as in Figure 3.3(b).

Example: Gaussian Elimination

Solve the system

$$\begin{cases} x + y + z = 4 \\ 2x + y - z = 1 \\ x + 2y + 3z = 7 \\ 3x + 2y = 5 \end{cases}$$

Solution. We reduce the augmented matrix:

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 1 & 2 & 3 & 7 \\ 3 & 2 & 0 & 5 \end{array}\right] \xrightarrow[R_4 - 3R_1 \to R_4]{\substack{R_2 - 2R_1 \to R_2 \\ R_3 - R_1 \to R_3}} \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & \boxed{-1} & -3 & -7 \\ 0 & 1 & 2 & 3 \\ 0 & -1 & -3 & -7 \end{array}\right] \xrightarrow{-R_2 \to R_2} \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & -1 & -3 & -7 \end{array}\right]$$

$$\xrightarrow[R_4+R_2\to R_4]{R_3-R_2\to R_3}\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & \boxed{-1} & -4 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-R_3\to R_3}\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Back-substitution: $z = 4 \implies y = 7 - 3(4) = -5 \implies x = 4 - (-5) - 4 = 5$.

Gauss–Jordan elimination continues from REF to RREF:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[R_1-R_3\to R_1]{R_2-3R_3\to R_2}\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_1-R_2\to R_1}\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Thus, $(x, y, z) = (5, -5, 4)$.

Example: Solving a 4×3 System with no Solution Using Gaussian Elimination

Solve the system

$$\begin{cases} x + y + z = 4 \\ 2x + y - z = 1 \\ x + 2y + 3z = 7 \\ x + 3y + 4z = 10 \end{cases}.$$

Solution. We reduce the augmented matrix:

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 1 & 2 & 3 & 7 \\ 1 & 3 & 4 & 10 \end{array}\right] \xrightarrow[R_4-R_1\to R_4]{\substack{R_2-2R_1\to R_2 \\ R_3-R_1\to R_3}}\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & \boxed{-1} & -3 & -7 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 3 & 6 \end{array}\right] \xrightarrow{-R_2\to R_2}\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 3 & 6 \end{array}\right]$$

$$\xrightarrow[R_4-2R_2\to R_4]{R_3-R_2\to R_3}\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & \boxed{-1} & -4 \\ 0 & 0 & -3 & -8 \end{array}\right] \xrightarrow{-R_3\to R_3}\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -3 & -8 \end{array}\right] \xrightarrow{R_4+3R_3\to R_4}\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 4 \end{array}\right].$$

The last row gives $0 = 4$, which is inconsistent. Thus, the system has **no solution**. *Geometric interpretation:* The first three planes intersect at the single point $(5, -5, 4)$, but the fourth plane does not pass through it ($5 - 15 + 16 = 6 \neq 10$).

Example: Solving a 3×4 System (Infinitely Many Solutions)

Solve the system

$$\begin{cases} x + y + z + w = 4 \\ 2x + y - z + w = 1 \\ x + 2y + 3z + w = 7 \end{cases}.$$

Solution. We reduce the augmented matrix:

$$\left[\begin{array}{cccc|c} \boxed{1} & 1 & 1 & 1 & 4 \\ 2 & 1 & -1 & 1 & 1 \\ 1 & 2 & 3 & 1 & 7 \end{array}\right] \xrightarrow[R_3-R_1\to R_3]{R_2-2R_1\to R_2}\sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & \boxed{-1} & -3 & -1 & -7 \\ 0 & 1 & 2 & 0 & 3 \end{array}\right] \xrightarrow{-R_2\to R_2}\sim \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 4 \\ 0 & 1 & 3 & 1 & 7 \\ 0 & 1 & 2 & 0 & 3 \end{array}\right]$$

$$\xrightarrow{R_3-R_2\to R_3}\sim\left[\begin{array}{cccc|c}1&1&1&1&4\\0&1&3&1&7\\0&0&\boxed{-1}&-1&-4\end{array}\right]\xrightarrow{-R_3\to R_3}\sim\left[\begin{array}{cccc|c}1&1&1&1&4\\0&1&3&1&7\\0&0&1&1&4\end{array}\right].$$

Column 4 has no pivot, so w is a free variable.

Back-substitution: Setting $w = t \in \mathbb{R}$:

$$\begin{aligned} z &= 4 - t, \\ y &= 7 - 3(4 - t) - t = -5 + 2t, \\ x &= 4 - (-5 + 2t) - (4 - t) - t = 5 - 2t. \end{aligned}$$

General Solution in vector form ($t \in \mathbb{R}$):

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} 5 \\ -5 \\ 4 \\ 0 \end{bmatrix}}_{\text{particular solution}} + t \underbrace{\begin{bmatrix} -2 \\ 2 \\ -1 \\ 1 \end{bmatrix}}_{\text{direction vector}}, \quad t \in \mathbb{R}.$$

The solution set is a line in $\mathbb{R}^4$.

Intuition. **Free variables are degrees of freedom.** Every pivot is one independent condition that determines one variable; the variables without a pivot are left unconstrained. The three equations of this example give only three conditions on four unknowns, so exactly one unknown (w) can be chosen freely and the others follow: the solutions form a line, a one-parameter family. In general the number of free variables is $n - \operatorname{rank}(A)$.

Common mistake. **Describing infinitely many solutions.** The answer must show *all* variables in terms of the parameters; "w is free" alone is not an answer. The parameters belong to the free variables (columns without a pivot), and a free variable is not necessarily the last one (see the example with a skipped column). Different parametrizations describe the same set: $(5 - 2t, -5 + 2t, 4 - t, t)$ and, with $t = 1 + u$, $(3 - 2u, -3 + 2u, 3 - u, 1 + u)$ are the same line.

3.2.4 Rank of a Matrix

Definition

The **rank** of a matrix A, denoted $\operatorname{rank}(A)$, is the number of nonzero rows in the **row-echelon form** (REF) of A (equivalently, the number of pivots), or the dimension of the row space (or column space) of A. For an $m \times n$ matrix, $0 \le \operatorname{rank}(A) \le \min(m, n)$.

The **Fundamental Theorem of Systems of Linear Equations** (also known as the **Kronecker–Capelli** or **Rouché–Capelli theorem**) states that for a system of linear equations $AX = B$, where A is an $m \times n$ coefficient matrix, X is an $n \times 1$ variable matrix, and B is an $m \times 1$ constant matrix:

- The system has a **solution** if and only if $\operatorname{rank}(A) = \operatorname{rank}([A|B])$.
- If $\operatorname{rank}(A) = \operatorname{rank}([A|B]) = n$, the system has a **unique solution**.
- If $\operatorname{rank}(A) = \operatorname{rank}([A|B]) = r < n$, the system has **infinitely many solutions** with $n - r$ free variables.
- If $\operatorname{rank}(A) < \operatorname{rank}([A|B])$, the system has **no solution** (inconsistent).

The decision procedure is summarized in Figure 3.8. (Note that always $\operatorname{rank}(A) \le \operatorname{rank}([A|B]) \le \operatorname{rank}(A) + 1$, so only these two cases can occur.)

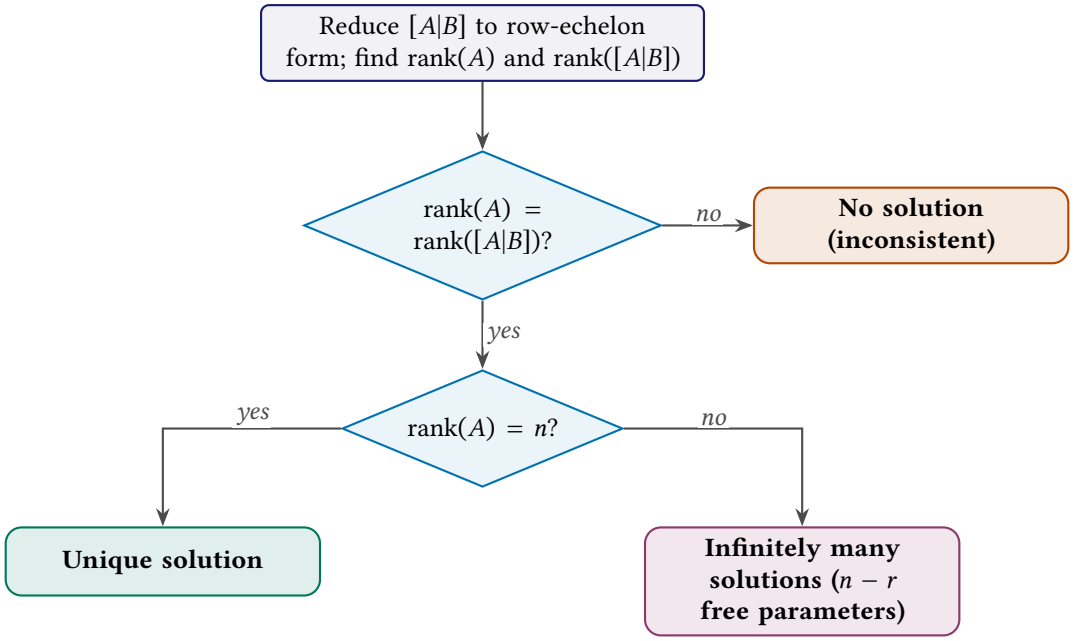


Figure 3.8: Classifying the system $AX = B$ (n = number of variables, $r = \operatorname{rank}(A)$).

Intuition. **Why does the rank criterion work?** $\operatorname{rank}(A)$ counts the independent columns of A. Adding the column B to form $[A|B]$ either leaves the rank unchanged (then B is built from the columns of A, i.e. the system has a solution: this is the column picture of Figure 3.5) or raises it by one (B points in a genuinely new direction and cannot be built: no solution). If $\operatorname{rank}(A) = n$ the columns are independent and there is only one way to build B; if $\operatorname{rank}(A) < n$ some column is a combination of the others and there are many ways, one free parameter for each missing rank.

Common mistake. **Compare $\operatorname{rank}(A)$ with $\operatorname{rank}([A|B])$, and with n.** The constants column counts when you decide whether a row is zero: the row $[\,0\;0\;0 \mid 7\,]$ is nonzero in $[A|B]$ but zero in A. And $\operatorname{rank}(A) = \operatorname{rank}([A|B])$ proves only that a solution *exists*; whether it is unique depends on comparing the common rank with the number of unknowns n.

Example: Finding the Rank of a 3×4 Matrix

Find the rank of the matrix

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & -1 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

by reducing it to row-echelon form.

Solution. We reduce the matrix to row-echelon form:

$$\begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 2 & 1 & -1 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \xrightarrow[R_3 - R_1 \to R_3]{R_2 - 2R_1 \to R_2} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & -3 & -1 \\ 0 & 1 & 2 & 0 \end{bmatrix} \xrightarrow{-R_2 \to R_2} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & \boxed{1} & 3 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

$$\xrightarrow{R_3 - R_2 \to R_3} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix} \xrightarrow{-R_3 \to R_3} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & \boxed{1} & 1 \end{bmatrix}.$$

The matrix is now in row-echelon form. Since there are 3 nonzero rows (pivots), $\operatorname{rank}(A) = 3$.

3.2.5 Homogeneous Systems

A **homogeneous system** of linear equations is a system of the form $AX = 0$, where A is an $m \times n$ **coefficient matrix**, X is the $n \times 1$ **variable matrix**, and 0 is the $m \times 1$ zero matrix. Key properties:

- A homogeneous system is always **consistent**, as $X = 0$ (the **trivial solution**) is a solution.
- If $\operatorname{rank}(A) = n$, the system has only the **trivial solution** ($X = 0$).
- If $\operatorname{rank}(A) < n$, the system has **infinitely many solutions**, including the trivial solution and **nontrivial solutions**, with $n - \operatorname{rank}(A)$ free variables.
- In particular, if there are fewer equations than unknowns ($m < n$), then $\operatorname{rank}(A) \le m < n$, so there is always a nontrivial solution.

Geometrically, every equation of a homogeneous system is a line (or plane) *through the origin*, so the origin always belongs to the solution set.

Why we need it. *Homogeneous systems describe "no input" situations (free vibrations, balanced forces, balancing chemical equations), and they hold the key to the structure of* all *solution sets (see the next box).*

Common mistake. **Homogeneous does not mean "only the trivial solution".** The zero solution always exists, but the interesting question is whether there are *others*: yes if and only if $\operatorname{rank}(A) < n$. Fewer equations than unknowns guarantees nontrivial solutions, but having at least as many equations as unknowns does *not* rule them out (dependent equations). For a square system the criterion is $\det A = 0$.

Example: A Homogeneous System

Solve

$$\begin{cases} x + y + z = 0 \\ x - y + 3z = 0 \end{cases}$$

Solution. Since $m = 2 < n = 3$, there is a nontrivial solution. The coefficient matrix (the right-hand side stays 0 throughout) is reduced as follows:

$$\begin{bmatrix} \boxed{1} & 1 & 1 \\ 1 & -1 & 3 \end{bmatrix} \xrightarrow{R_2 - R_1 \to R_2} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & \boxed{-2} & 2 \end{bmatrix} \xrightarrow{-\frac{1}{2}R_2 \to R_2} \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}.$$

The pivots are in columns 1 and 2, so z is free. Let $z = t$. Then $y = t$ and $x = -y - z = -2t$. The solutions are

$$(x, y, z) = (-2t,\ t,\ t) = t(-2,\ 1,\ 1), \quad t \in \mathbb{R}.$$

The two planes pass through the origin and meet in the line through the origin with direction $(-2, 1, 1)$.

Intuition. **Structure of the solution set: particular plus homogeneous.** If X_p is *one* solution of $AX = B$ (a *particular solution*), then every solution has the form

$$X = X_p + X_h, \qquad \text{where } AX_h = 0,$$

that is, the general solution of $AX = B$ equals a particular solution plus the general solution of the associated homogeneous system $AX = 0$. Indeed, $A(X_p + X_h) = B + 0 = B$, and conversely $A(X - X_p) = B - B = 0$. In the 3×4 example above, $X_p = (5, -5, 4, 0)$ and $X_h = t(-2, 2, -1, 1)$. This is why the solution sets of consistent systems look like lines, planes, ... that do not necessarily pass through the origin (Figure 3.9).

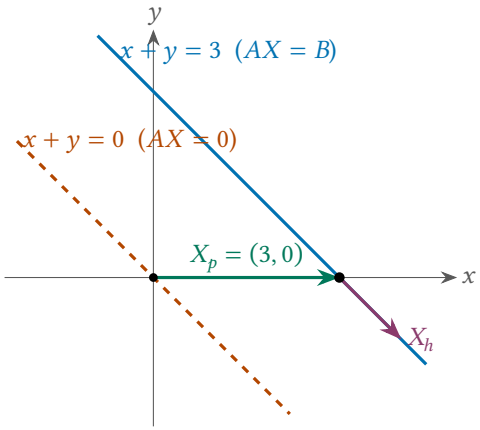


Figure 3.9: For the system $x + y = 3$ (one equation, two unknowns): the homogeneous system $x + y = 0$ has the solution line $\{t(1,-1)\}$ through the origin (dashed). Adding one particular solution $X_p = (3,0)$ *shifts* this line onto the solution line of $x + y = 3$: the two solution sets are parallel.

3.3 Systems of Linear Equations in Physics and Engineering

Whenever several unknown quantities are linked by balance laws, the model is a linear system: Kirchhoff's laws in circuits, force equilibrium in structures, conservation of cars, water or heat in networks, conservation of atoms in chemistry, and fitting a curve through measured points. The mathematics of this chapter then has a direct physical meaning. A *free variable* is a quantity that the available laws cannot fix (a design freedom or a missing measurement); an *inconsistent* system signals contradictory data or a wrongly stated model; a *unique* solution means that the laws determine everything.

3.3.1 Electric Circuits: Mesh Currents

Kirchhoff's voltage law says that the voltage rises and drops around every closed loop add up to zero. Assign a clockwise *mesh current* to each loop; a resistor shared by two loops carries the *difference* of the two mesh currents. A circuit with ℓ independent loops gives an $\ell \times \ell$ linear system for the currents.

Example: A Two-Loop Circuit

In Figure 3.10(a) two batteries (12 V and 6 V) and three resistors ($R_1 = 2\,\Omega$, $R_2 = 3\,\Omega$, shared $R_3 = 4\,\Omega$) form two loops. The shared resistor carries $I_1 - I_2$, so the voltage law gives

$$\begin{cases} 2I_1 + 4(I_1 - I_2) = 12 \\ 3I_2 + 4(I_2 - I_1) = 6 \end{cases} \quad\Longleftrightarrow\quad \begin{cases} 6I_1 - 4I_2 = 12 \\ -4I_1 + 7I_2 = 6. \end{cases}$$

Here $\det A = 42 - 16 = 26 \neq 0$, and Cramer's rule gives

$$I_1 = \frac{\begin{vmatrix} 12 & -4 \\ 6 & 7 \end{vmatrix}}{26} = \frac{108}{26} \approx 4.15\ \text{A}, \qquad I_2 = \frac{\begin{vmatrix} 6 & 12 \\ -4 & 6 \end{vmatrix}}{26} = \frac{84}{26} \approx 3.23\ \text{A}.$$

The current in the shared resistor is $I_1 - I_2 = \frac{12}{13} \approx 0.92$ A.

Check (energy balance). The batteries deliver $12 \cdot \frac{54}{13} + 6 \cdot \frac{42}{13} = \frac{900}{13} \approx 69.2$ W, and the resistors dissipate $2I_1^2 + 3I_2^2 + 4(I_1 - I_2)^2 = \frac{900}{13}$ W, the same.

Intuition. **Why this system always has a unique solution.** For a network of positive resistors the matrix of the mesh equations is always invertible. Physically this says that the currents are completely determined by the sources: a circuit cannot "choose" between two different current patterns. If such a system ever turns out singular, the circuit has been drawn incorrectly (for example a loop of ideal wires and ideal sources).

3.3.2 Statics: Tensions in Cables

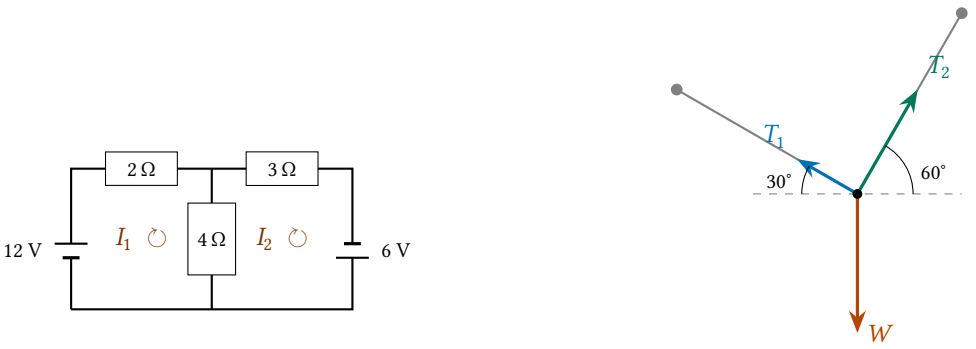
A body at rest is in equilibrium: the sum of all forces on it is zero, component by component. Each component is one linear equation in the unknown forces, even when the coefficients are sines and cosines of fixed angles (the unknowns are the tensions, not the angles).

Example: A Hanging Load

A load of weight $W = 100$ N hangs from two cables that make angles 30° (left) and 60° (right) with the horizontal (Figure 3.10(b)). Let T_1, T_2 be the tensions. Horizontal and vertical equilibrium give

$$\begin{cases} -T_1 \cos 30^\circ + T_2 \cos 60^\circ = 0 \\ T_1 \sin 30^\circ + T_2 \sin 60^\circ = 100 \end{cases} \iff \begin{cases} -\frac{\sqrt{3}}{2} T_1 + \frac{1}{2} T_2 = 0 \\ \frac{1}{2} T_1 + \frac{\sqrt{3}}{2} T_2 = 100. \end{cases}$$

From the first equation $T_2 = \sqrt{3}\, T_1$; substituting, $\frac{1}{2}T_1 + \frac{3}{2}T_1 = 100$, so $T_1 = 50$ N and $T_2 = 50\sqrt{3} \approx 86.6$ N. Check: the vertical components are $50 \cdot \frac{1}{2} + 86.6 \cdot \frac{\sqrt{3}}{2} = 25 + 75 = 100$ N.



(a) two mesh currents

(b) forces at the knot

Figure 3.10: (a) A two-loop circuit; both batteries drive their mesh current clockwise (the long plate is the positive terminal), and the middle resistor carries $I_1 - I_2$. (b) Free-body diagram of the knot where the load hangs: three forces in equilibrium, $T_1 = 50$ N, $T_2 \approx 86.6$ N, $W = 100$ N (arrow lengths are drawn to scale).

Common mistake. **Coefficients may be trigonometric, but the system is still linear.** In statics the unknowns are *forces*; the angles are given numbers. Do not treat $\cos 30^\circ$ as a variable. Also, a structure with *more* unknown forces than independent equilibrium equations gives a system with free variables: such a structure is called *statically indeterminate*. Equilibrium alone cannot fix the forces; one must add the elasticity of the members.

3.3.3 Chemistry: Balancing a Reaction

A chemical equation is balanced when every kind of atom occurs equally often on both sides. For unknown coefficients $x_1, x_2, \ldots$ this is a *homogeneous* linear system (right-hand sides zero), and since only the ratios matter, we expect a nontrivial solution with a free parameter.

Example: Burning Ethane

Balance $x_1\,C_2H_6 + x_2\,O_2 \to x_3\,CO_2 + x_4\,H_2O$. Counting atoms of each kind:

$$\begin{cases} \text{C:} & 2x_1 = x_3 \\ \text{H:} & 6x_1 = 2x_4 \\ \text{O:} & 2x_2 = 2x_3 + x_4 \end{cases} \qquad \left[\begin{array}{cccc|c} 2 & 0 & -1 & 0 & 0 \\ 6 & 0 & 0 & -2 & 0 \\ 0 & 2 & -2 & -1 & 0 \end{array}\right].$$

Three equations in four unknowns: $\operatorname{rank}(A) = 3 < 4$, so there is one free variable. Let $x_1 = t$. Then $x_3 = 2t$, $x_4 = 3t$ and $x_2 = \frac{2x_3 + x_4}{2} = \frac{7}{2}t$. The smallest positive *integer* solution is $t = 2$:

$$2\,C_2H_6 + 7\,O_2 \to 4\,CO_2 + 6\,H_2O.$$

The free parameter is the size of the batch: doubling every coefficient gives the same reaction.

3.3.4 Flow Networks: Traffic and Pipes

At every junction of a network the flow in equals the flow out (conservation of cars, water, current, ...). Each junction gives one linear equation for the unknown flows on the roads or pipes.

Example: A Traffic Ring

Four one-way streets form a ring (Figure 3.11(a)); $x_1, \ldots, x_4$ are the flows (cars per hour) on the streets AB, BC, CD, DA, and cars enter and leave at the junctions as drawn. "In = out" at each junction:

$$\begin{cases} A: & x_4 + 300 = x_1 + 100 \\ B: & x_1 + 200 = x_2 + 300 \\ C: & x_2 + 150 = x_3 + 250 \\ D: & x_3 = x_4 \end{cases} \iff \begin{cases} x_1 - x_4 = 200 \\ x_1 - x_2 = 100 \\ x_2 - x_3 = 100 \\ x_3 - x_4 = 0. \end{cases}$$

The equations are consistent (the last three add up to $x_1 - x_4 = 200$), but only three of them are independent, so one variable is free. With $x_4 = t$:

$$(x_1, x_2, x_3, x_4) = (t + 200,\ t + 100,\ t,\ t), \qquad t \geq 0.$$

The conservation law cannot tell how many cars circulate around the ring (t). One traffic count on any single street fixes the whole pattern: if 80 cars per hour are counted on DA, then $(x_1, x_2, x_3, x_4) = (280, 180, 80, 80)$.

3.3.5 Curve Fitting: Polynomial Interpolation

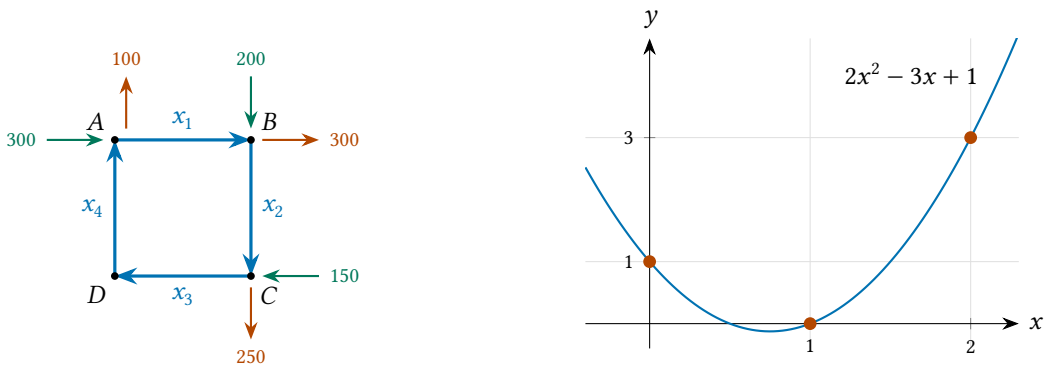
To find the parabola $p(x) = ax^2 + bx + c$ through three measured points, require $p(x_i) = y_i$. Each point gives one equation, *linear in the unknown coefficients* a, b, c (the numbers $x_i^2, x_i, 1$ are the coefficients).

Example: A Parabola Through Three Points

Fit $p(x) = ax^2 + bx + c$ through $(0,1)$, $(1,0)$, $(2,3)$:

$$\begin{cases} c = 1 \\ a + b + c = 0 \\ 4a + 2b + c = 3 \end{cases} \implies \begin{cases} a + b = -1 \\ 4a + 2b = 2 \end{cases} \implies a = 2,\ b = -3,\ c = 1.$$

So $p(x) = 2x^2 - 3x + 1$ (Figure 3.11(b)). The coefficient matrix is a *Vandermonde matrix*, with $\det = (1-0)(2-0)(2-1) = 2 \neq 0$, which is why there is exactly one parabola; it would be singular if two of the x-values coincided.



(a) traffic ring (green: in, red: out)

(b) the parabola through three points

Figure 3.11: (a) Conservation of cars at the four junctions of a ring road leaves one free parameter, the circulating traffic t. (b) Three points determine exactly one parabola; the three equations for its coefficients form a system with a unique solution.

3.3.6 Heat Conduction: a Large, Sparse System

In steady state, a point inside a uniform rod has the *average* temperature of its neighbors. Writing this at every grid point gives a linear system whose size equals the number of grid points. Real models of heat flow, stress or fluid flow use thousands to millions of grid points, so the *size* of the system, not its difficulty, is the challenge: the methods of this chapter must be efficient, and they exploit the fact that almost all coefficients are zero.

Example: A Rod with Fixed End Temperatures

A rod is held at 100°C at the left end and 0°C at the right end and divided into four equal parts. The temperatures T_1, T_2, T_3 at the three interior points satisfy "each is the average of its neighbors":

$$\begin{cases} 2T_1 - T_2 = 100 \\ -T_1 + 2T_2 - T_3 = 0 \\ -T_2 + 2T_3 = 0 \end{cases} \implies T_3 = \tfrac{1}{2}T_2, \quad T_1 = \tfrac{3}{2}T_2, \quad 2 \cdot \tfrac{3}{2}T_2 - T_2 = 100,$$

so $T_2 = 50$, $T_1 = 75$, $T_3 = 25$: the temperature falls linearly from 100°C to 0°C (Figure 3.12). With n interior points the matrix is $n \times n$ with 2 on the diagonal and -1 next to it, and it is always invertible, so the temperature profile is unique.

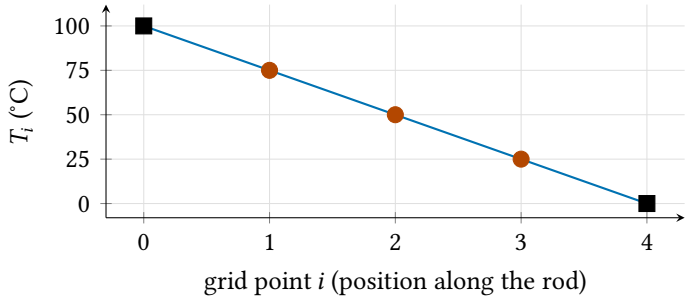


Figure 3.12: Steady-state temperatures along the rod. The squares are the prescribed end temperatures; the dots are the unknowns T_1, T_2, T_3 obtained from the linear system.

Common mistake. A mathematical solution need not be a physical one. Flows, amounts of food, numbers of widgets and reaction coefficients must be non-negative (often integers). For the traffic ring this means $t \geq 0$; for the widgets of the homework it leaves a single solution among infinitely many real ones. Always substitute back and ask: does this make sense in the real problem? And with *measured* data, a system with more equations than unknowns is almost never exactly consistent; then one looks for the best compromise (least squares) instead of an exact solution.

3.4 Facts and Curiosities

1. Two thousand years before Gauss. The Chinese text *The Nine Chapters on the Mathematical Art* (compiled by about the first century CE, with older material) solves systems of linear equations by arranging the numbers in a rectangular array of counting rods and eliminating step by step: essentially what we now call Gaussian elimination. Gauss used the method in 1809 for orbit calculations, and the name stuck.

2. Who is Jordan? The "Jordan" of Gauss–Jordan elimination is Wilhelm Jordan (1842–1899), a German geodesist who described the method in his handbook of surveying; he is often confused with the mathematician Camille Jordan. Cramer published his rule in 1750 (Maclaurin had found it earlier). The rank criterion carries several names (Rouché–Capelli in the West, Kronecker–Capelli in the East).

3. Why exactly three outcomes? Linearity is the reason: the line through two solutions consists of solutions, so two solutions force infinitely many. A nonlinear equation behaves differently: $x^2 = 1$ has exactly two solutions.

4. The cost of a method. Gaussian elimination needs about $\frac{1}{3}n^3$ multiplications: for $n = 1000$ unknowns that is about $3.3 \cdot 10^8$, a fraction of a second on a laptop. Cramer's rule with cofactor expansion needs determinants that take of the order of $n!$ operations: for $n = 20$ already $2.4 \cdot 10^{18}$, about 77 years at 10^9 operations per second. This is why software always uses elimination (LU factorization) and Cramer's rule is a tool for small systems and for formulas.

5. Generic and special systems. If the coefficients of a square system are chosen at random, the system almost surely has a unique solution: "no solution" and "infinitely many" are exceptional. Yet they appear constantly in practice, because real models have structure (symmetry, conservation laws, repeated equations), and that structure is exactly what makes the matrix singular. The traffic ring is an example: conservation makes one equation redundant.

6. Nearly parallel lines are dangerous. Two lines that are almost parallel meet at a point that moves enormously when one line is shifted slightly (Figure 3.13). Systems of this kind are called *ill-conditioned*: a tiny error in the data (a measurement, a rounding) produces a large error in the solution. Mathematically the solution is unique, but numerically it is unreliable.

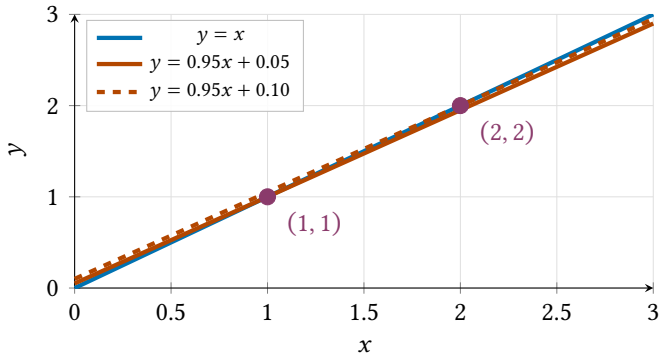


Figure 3.13: An ill-conditioned system. Shifting the second line up by only 0.05 moves the solution from $(1, 1)$ to $(2, 2)$: a shift of the data by 5% of a unit moves the answer by a whole unit, because the two lines are almost parallel.

7. Elimination works over other number systems. All the algebra of this chapter uses only $+, -, \times, \div$, so it works unchanged over the rationals, the complex numbers, or the integers modulo 2. Linear systems modulo 2 are used to decode error-correcting codes (the way a damaged message on a disc or in a data link is repaired), and rounding errors in floating-point arithmetic are the reason why numerical pivoting strategies exist.

3.5 Worked Problems

Problem: Solving a System Using Cramer's Rule

Solve the system

$$\begin{cases} 2x + y - z = 3 \\ x + 3y + 2z = 7 \\ 3x - y + z = 5 \end{cases}$$

using Cramer's Rule.

Solution. Write as $AX = B$, where,

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 3 & 2 \\ 3 & -1 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 7 \\ 5 \end{bmatrix}.$$

1. Compute $\det(A) = 2(3 \cdot 1 - 2 \cdot (-1)) - 1(1 \cdot 1 - 2 \cdot 3) + (-1)(1 \cdot (-1) - 3 \cdot 3) = 25$.
2. For x, form $A_1 = \begin{bmatrix} 3 & 1 & -1 \\ 7 & 3 & 2 \\ 5 & -1 & 1 \end{bmatrix}$. $\det(A_1) = 3(3 \cdot 1 - 2 \cdot (-1)) - 1(7 \cdot 1 - 2 \cdot 5) + (-1)(7 \cdot (-1) - 3 \cdot 5) = 40$.

   Thus, $x = \dfrac{\det(A_1)}{\det(A)} = \dfrac{40}{25} = \dfrac{8}{5}$.

3. For y, form $A_2 = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 7 & 2 \\ 3 & 5 & 1 \end{bmatrix}$. $\det(A_2) = 2(7\cdot 1 - 2\cdot 5) - 3(1\cdot 1 - 2\cdot 3) + (-1)(1\cdot 5 - 7\cdot 3) = 25$.
   Thus, $y = \frac{\det(A_2)}{\det(A)} = \frac{25}{25} = 1$.
4. For z, form $A_3 = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 3 & 7 \\ 3 & -1 & 5 \end{bmatrix}$. $\det(A_3) = 2(3\cdot 5 - 7\cdot(-1)) - 1(1\cdot 5 - 7\cdot 3) + 3(1\cdot(-1) - 3\cdot 3) = 30$.
   Thus, $z = \frac{\det(A_3)}{\det(A)} = \frac{30}{25} = \frac{6}{5}$.

Problem: Solving a System Using Gaussian Elimination

Solve the system

$$\begin{cases} 2x + y - z = 3 \\ x + 3y + 2z = 7 \\ 3x - y + z = 5 \end{cases}$$

using Gaussian Elimination (compare with the previous problem: same system, different method).

Solution. We reduce the augmented matrix:

$$\left[\begin{array}{ccc|c} \boxed{2} & 1 & -1 & 3 \\ 1 & 3 & 2 & 7 \\ 3 & -1 & 1 & 5 \end{array}\right] \xrightarrow{\frac{1}{2}R_1 \to R_1} \sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{3}{2} \\ 1 & 3 & 2 & 7 \\ 3 & -1 & 1 & 5 \end{array}\right] \xrightarrow[R_3 - 3R_1 \to R_3]{R_2 - R_1 \to R_2} \sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{3}{2} \\ 0 & \frac{5}{2} & \frac{5}{2} & \frac{11}{2} \\ 0 & -\frac{5}{2} & \frac{5}{2} & \frac{1}{2} \end{array}\right]$$

$$\xrightarrow{\frac{2}{5}R_2 \to R_2} \sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{3}{2} \\ 0 & \boxed{1} & 1 & \frac{11}{5} \\ 0 & -\frac{5}{2} & \frac{5}{2} & \frac{1}{2} \end{array}\right] \xrightarrow{R_3 + \frac{5}{2}R_2 \to R_3} \sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 1 & \frac{11}{5} \\ 0 & 0 & \boxed{5} & 6 \end{array}\right] \xrightarrow{\frac{1}{5}R_3 \to R_3} \sim \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 1 & \frac{11}{5} \\ 0 & 0 & 1 & \frac{6}{5} \end{array}\right].$$

Back-substitution: $z = \frac{6}{5} \implies y = \frac{11}{5} - \frac{6}{5} = 1 \implies x = \frac{3}{2} - \frac{1}{2}(1) + \frac{1}{2}\left(\frac{6}{5}\right) = \frac{8}{5}$. Thus, $(x, y, z) = \left(\frac{8}{5}, 1, \frac{6}{5}\right)$.

Problem: Gaussian Elimination Producing a Contradictory Row (No Solution)

Solve the system

$$\begin{cases} x + y + z = 1 \\ 2x + y + z = 2 \\ x + 2y + z = 4 \\ x + y + 2z = 5 \end{cases}$$

using Gaussian Elimination.

Solution. We reduce the augmented matrix:

$$\left[\begin{array}{ccc|c} \boxed{1} & 1 & 1 & 1 \\ 2 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & 2 & 5 \end{array}\right] \xrightarrow[R_4 - R_1 \to R_4]{\substack{R_2 - 2R_1 \to R_2 \\ R_3 - R_1 \to R_3}} \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & \boxed{-1} & -1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right] \xrightarrow{-R_2 \to R_2} \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

$$\xrightarrow{R_3-R_2\to R_3}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&1&0\\0&0&\boxed{-1}&3\\0&0&1&4\end{array}\right]\xrightarrow{-R_3\to R_3}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&1&0\\0&0&1&-3\\0&0&1&4\end{array}\right]\xrightarrow{R_4-R_3\to R_4}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&1&0\\0&0&1&-3\\0&0&0&7\end{array}\right].$$

The last row $[0\,0\,0\,|\,7]$ represents $0=7$, which is a contradiction. Thus, the system has **no solution**.

Intuition. Why no column was skipped here. In this example no column had to be skipped, because a pivot was found in each of the columns 1, 2 and 3. The rank of the coefficient matrix is 3, the rank of the augmented matrix is 4. Skipping a column happens when the pivot position and everything below it are zero; see the example "A Zero Pivot, Skipping a Column" in Section 3.2.3.

Problem: Determining Solution Type Using the Fundamental Theorem

Consider the system

$$\begin{cases}x+y+z=1\\2x+y-z=2\\x+2y+3z=4\\3x+2y+z=3\end{cases}$$

Determine the type of solution using the Fundamental Theorem of Systems of Linear Equations by computing $\operatorname{rank}(A)$ and $\operatorname{rank}([A|B])$.

Solution. We reduce $[A|B]$ to row-echelon form:

$$\left[\begin{array}{ccc|c}1&1&1&1\\2&1&-1&2\\1&2&3&4\\3&2&1&3\end{array}\right]\xrightarrow[R_4-3R_1\to R_4]{\substack{R_2-2R_1\to R_2\\R_3-R_1\to R_3}}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&-1&-3&0\\0&1&2&3\\0&-1&-2&0\end{array}\right]\xrightarrow{-R_2\to R_2}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&3&0\\0&1&2&3\\0&-1&-2&0\end{array}\right]$$

$$\xrightarrow[R_4+R_2\to R_4]{R_3-R_2\to R_3}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&3&0\\0&0&-1&3\\0&0&1&0\end{array}\right]\xrightarrow{-R_3\to R_3}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&3&0\\0&0&1&-3\\0&0&1&0\end{array}\right]\xrightarrow{R_4-R_3\to R_4}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&3&0\\0&0&1&-3\\0&0&0&3\end{array}\right].$$

Thus, $\operatorname{rank}([A|B])=4$ and $\operatorname{rank}(A)=3$. Since $\operatorname{rank}(A)=3<\operatorname{rank}([A|B])=4$, the system has **no solution** by the **Rouché–Capelli Theorem**.

Problem: A System with a Parameter

For which values of a and b does the system

$$\begin{cases}x+y+z=1\\x+2y+3z=2\\2x+3y+az=b\end{cases}$$

have (i) a unique solution, (ii) no solution, (iii) infinitely many solutions?

Solution. Reduce the augmented matrix:

$$\left[\begin{array}{ccc|c}1&1&1&1\\1&2&3&2\\2&3&a&b\end{array}\right]\xrightarrow[R_3-2R_1\to R_3]{R_2-R_1\to R_2}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&2&1\\0&1&a-2&b-2\end{array}\right]\xrightarrow{R_3-R_2\to R_3}\sim\left[\begin{array}{ccc|c}1&1&1&1\\0&1&2&1\\0&0&a-4&b-3\end{array}\right].$$

The behavior depends on the last row $[\,0 \;\; 0 \;\; a-4 \mid b-3\,]$:

- If $a \neq 4$: the last row has a pivot in column 3, so $\operatorname{rank}(A) = \operatorname{rank}([A|B]) = 3 = n$ and the solution is **unique** (for every b).
- If $a = 4$ and $b \neq 3$: the last row is $[0\ 0\ 0 \mid b-3]$ with $b-3 \neq 0$, so $\operatorname{rank}(A) = 2 < \operatorname{rank}([A|B]) = 3$: **no solution**.
- If $a = 4$ and $b = 3$: the last row is zero, $\operatorname{rank}(A) = \operatorname{rank}([A|B]) = 2 < 3$: **infinitely many solutions** with one free parameter ($z = t$, $y = 1 - 2t$, $x = t$).

Problem: Two Free Parameters and the Vector Form

Solve the system

$$\begin{cases} x + 2y - z + w = 3 \\ 2x + 4y - z + 3w = 8 \end{cases}$$

and write the solution in vector form.

Solution. The augmented matrix is $\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 3 \\ 2 & 4 & -1 & 3 & 8 \end{array}\right]$. The operation $R_2 - 2R_1 \to R_2$ gives

$$\left[\begin{array}{cccc|c} \boxed{1} & 2 & -1 & 1 & 3 \\ 0 & 0 & \boxed{1} & 1 & 2 \end{array}\right].$$

Column 2 has no pivot (its entry in row 2 is 0), so we skip it and find the second pivot in column 3. The pivots are in columns 1 and 3: x, z are pivot variables and y, w are free. Put $y = s$, $w = t$. Row 2 gives $z = 2 - t$, and row 1 gives $x = 3 - 2s + z - w = 3 - 2s + (2 - t) - t = 5 - 2s - 2t$. In vector form

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 2 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \qquad s, t \in \mathbb{R}.$$

Here $\operatorname{rank}(A) = 2$ and $n = 4$, so $n - \operatorname{rank}(A) = 2$ free parameters: the solution set is a *plane* in $\mathbb{R}^4$. (Check with $s = t = 0$: $5 - 2 = 3$ and $10 - 2 = 8$.)

Problem: A Consistency Condition

For which right-hand sides (b_1, b_2, b_3) is the system $x + 2y = b_1$, $2x + 4y = b_2$, $3x + 6y = b_3$ solvable? Describe the solutions.

Solution. Reduce the augmented matrix: $R_2 - 2R_1 \to R_2$, $R_3 - 3R_1 \to R_3$ give

$$\left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & 0 & b_2 - 2b_1 \\ 0 & 0 & b_3 - 3b_1 \end{array}\right].$$

The system is consistent if and only if $b_2 - 2b_1 = 0$ and $b_3 - 3b_1 = 0$, that is, if $B = b_1(1, 2, 3)$ is a multiple of the first column. Then $\operatorname{rank}(A) = \operatorname{rank}([A|B]) = 1 < 2$ and the solutions are $x = b_1 - 2y$ with y free. In the column picture: the two columns $(1, 2, 3)$ and $(2, 4, 6)$ are parallel, so they can only build vectors on one line through the origin.

Problem: A Homogeneous System and a Parameter

For which λ does the system

$$\begin{cases} x + 2y - z = 0 \\ 2x + 3y + \lambda z = 0 \\ x + y + 2z = 0 \end{cases}$$

have nontrivial solutions? Find them.

Solution. A square homogeneous system has nontrivial solutions exactly when $\det A = 0$. Expanding,

$$\det \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & \lambda \\ 1 & 1 & 2 \end{bmatrix} = 1(6-\lambda) - 2(4-\lambda) + (-1)(2-3) = \lambda - 1,$$

so $\lambda = 1$. For $\lambda = 1$: $R_2 - 2R_1 \to R_2$ and $R_3 - R_1 \to R_3$ give two identical rows $(0, -1, 3)$, hence a zero row after $R_3 - R_2 \to R_3$. So $-y + 3z = 0$, $y = 3z$, and $x = -2y + z = -5z$:

$$(x, y, z) = t(-5,\ 3,\ 1), \qquad t \in \mathbb{R}.$$

For every other λ only the trivial solution $(0, 0, 0)$ exists.

3.6 Conceptual & Computational Exercises

3.6.1 Conceptual Questions

Answer by *thinking*, with little or no calculation; give a reason or a counterexample.

1. **True or false?**
   (a) A system of three equations in two unknowns never has a solution.
   (b) A system with more unknowns than equations always has infinitely many solutions.
   (c) A homogeneous system can be inconsistent.
   (d) If $\operatorname{rank}(A) = \operatorname{rank}([A|B])$, the system has exactly one solution.
   (e) If a system has two different solutions, it has infinitely many.
   (f) Interchanging two equations can change the solution set.
   (g) A square system with $\det A = 0$ never has a solution.
2. Why can a linear system never have exactly two solutions? Why can the equation $x^2 = 1$?
3. Describe all possible ways three planes in space can intersect (in a point, a line, a plane, or not at all) and give an example of each.
4. In the *row picture* a 2×2 system is a pair of lines. What is it in the *column picture*? When is there no solution in each picture?
5. The columns of A are $(1, 2)$ and $(2, 4)$. For which right-hand sides B does $AX = B$ have a solution? How many solutions are there when it does?
6. Why is it forbidden to multiply a row by 0 during elimination, while multiplying by -7 is allowed?
7. A 3×3 homogeneous system $AX = 0$ has $\det A \neq 0$. What is its solution set? What if $\det A = 0$?
8. Can a system of 4 equations in 3 unknowns have exactly one solution? Can it have infinitely many?
9. A consistent system has 2 equations and 3 unknowns. How many free parameters can its solution have?
10. After reducing a system you meet the row $[\,0\ 0\ 0 \mid 0\,]$. Is this an error? What if it is $[\,0\ 0\ 0 \mid 5\,]$?

11. Why do both the inverse method and Cramer's rule fail when $\det A = 0$? What can you do instead?
12. The solution sets of $AX = B$ and $AX = 0$ are related. How? Illustrate with $x + y = 3$ and $x + y = 0$.
13. A structural analysis produces a system with a free variable. What does this mean physically?
14. In a circuit with positive resistors the mesh-current matrix is always invertible. What physical fact does this express?

3.6.2 Computational Exercises

1. Solve the following system using Cramer's rule:

$$\begin{cases} x + 2y - z = 3 \\ 2x - y + 3z = 7 \\ x + y + z = 4 \end{cases}$$

2. Solve each system using Gaussian elimination, and describe the solution set geometrically (in which position are the planes?):

$$\begin{cases} x + y + z = 6 \\ 2x + 3y + z = 10 \\ -x + y + 2z = 3 \end{cases}, \quad \begin{cases} x + y + z = 4 \\ 2x + y - z = 1 \\ x + 2y + 3z = 7 \\ 2x + 3y + 4z = 10 \end{cases}, \quad \begin{cases} x + 2y - z = 3 \\ 2x + 4y - 2z = 6 \\ -x - 2y + z = -3 \end{cases}.$$

3. For the system

$$\begin{cases} x + y + z = 2 \\ 2x + 3y + z = 5 \\ x + 2y + 2z = 4 \end{cases}$$

determine whether the system has a **unique solution**, **no solution**, or **infinitely many solutions**, and find all solutions.

4. A factory produces three types of widgets: A, B, and C. Each A uses 2 units of material X and 1 unit of material Y. Each B uses 1 unit of X and 2 units of Y. Each C uses 3 units of X and 1 unit of Y.
   The factory has 10 units of X and 8 units of Y, and all the material is used up. Set up a system of linear equations, find all its solutions, and decide how many widgets of each type can be produced if the numbers of widgets must be non-negative integers.
5. Use a system of linear equations to determine the quantities of different foods needed to meet specific nutritional goals.
   Problem: You need to design a simple, single-meal diet plan that meets three nutritional targets: calories, protein, and carbohydrates. You have three food options:
   - **Grilled Chicken:** 200 calories, 40g of protein, 0g of carbohydrates per serving.
   - **Quinoa:** 150 calories, 5g of protein, 25g of carbohydrates per serving.
   - **Avocado:** 250 calories, 3g of protein, 13g of carbohydrates per serving.

   Your goal is to create a meal that has a total of 700 calories, 50g of protein, and 40g of carbohydrates. Let x, y, and z denote the number of servings of Grilled Chicken, Quinoa, and Avocado, respectively. Then the system of equations is:

$$\begin{cases} 200x + 150y + 250z = 700 & \text{(Calories)} \\ 40x + 5y + 3z = 50 & \text{(Protein)} \\ 0x + 25y + 13z = 40 & \text{(Carbohydrates)} \end{cases}$$

   Solve for x, y, and z to determine how many servings of each food are needed.

6. For which values of k does the system

$$\begin{cases} x + y + kz = 1 \\ x + ky + z = 1 \\ kx + y + z = 1 \end{cases}$$

have (a) a unique solution, (b) no solution, (c) infinitely many solutions? Use the rank criterion (Kronecker–Capelli) or Cramer's rule.

7. Solve the system and write the solution in vector form:

$$\begin{cases} x + y + 2z - w = 1 \\ 2x + 2y + 5z + w = 7 \end{cases}$$

8. For which (b_1, b_2, b_3) is the system $x - y = b_1,\ 2x + y = b_2,\ 3x = b_3$ solvable? When it is, solve it. Explain the condition in the column picture.

9. For which λ does $\begin{cases} x + 2y + z = 0 \\ 2x + \lambda y + 3z = 0 \\ x + y + 2z = 0 \end{cases}$ have nontrivial solutions? Find all solutions for that λ.

10. **(Chemistry)** Balance (a) $C_3H_8 + O_2 \to CO_2 + H_2O$ and (b) $Fe_2O_3 + C \to Fe + CO_2$ by setting up a homogeneous linear system. Give the smallest positive integer coefficients.

11. **(Circuit)** The mesh equations of a two-loop circuit are $5I_1 - 2I_2 = 9$ and $-2I_1 + 4I_2 = 0$ (currents in amperes). Solve them by Cramer's rule and find the current in the resistor shared by the two loops. Show that $\det A \neq 0$.

12. **(Statics)** A load of 200 N hangs from two cables. Find the tension in each cable (a) if both cables make 45° with the horizontal, (b) if both make 30°. What happens to the tensions as the cables become flatter?

13. **(Interpolation)** Find the parabola $y = ax^2 + bx + c$ through $(0, 2)$, $(1, 3)$, $(2, 8)$. Why is there no function $y = ax^2 + bx + c$ through $(1, 2)$, $(1, 5)$, $(2, 3)$?

14. **(Traffic)** In the traffic ring of this chapter a counter on street DA shows 80 cars per hour. Find all four flows. What is the smallest possible flow on street AB if no counter is available?